

Nametable Mapping

File: nametable_mapping.kicad_sch

CPU Address Bus

File: cpuAddressBus.kicad_sch

Register Access

File: register_access.kicad_sch

PRG Banking

File: prg_banking.kicad_sch

PPU Address Bus

File: ppu_address_bus.kicad_sch

Reset Detection

File: reset_detection.kicad_sch

CPU Data Bus

File: cpu_data_bus.kicad_sch

PCM DAC

File: pcm_dac.kicad_sch

Scanline Detection

File: scanline_detection.kicad_sch

V-Split

File: v_split.kicad_sch

PPU Fetch Counting

File: ppu_fetch_counting.kicad_sch

CHR A0,A1,A2

File: chr_a0_a1_a2.kicad_sch

ExRAM Mode

File: exram_mode.kicad_sch

Fill Mode

File: fill_mode.kicad_sch

Pulse Length Lookup Table

File: pulse_length_lookup_table.kicad_sch

Pulse Timer

File: pulse_timer.kicad_sch

Pulse Duty Envelope Volume

File: pulse_duty_envelope_volume.kicad_sch

Pulse DAC

File: Pulse DAC.kicad_sch

PPU Data Bus

File: ppu_data_bus.kicad_sch

CHR Bank Registers

File: chr_bank_registers.kicad_sch

CHR Bank Selection Logic

File: chr_bank_selection_logic.kicad_sch

CHR Address Bus 10 to 14

File: chr_address_bus_10_to_14.kicad_sch

CHR Address Bus 15 to 19

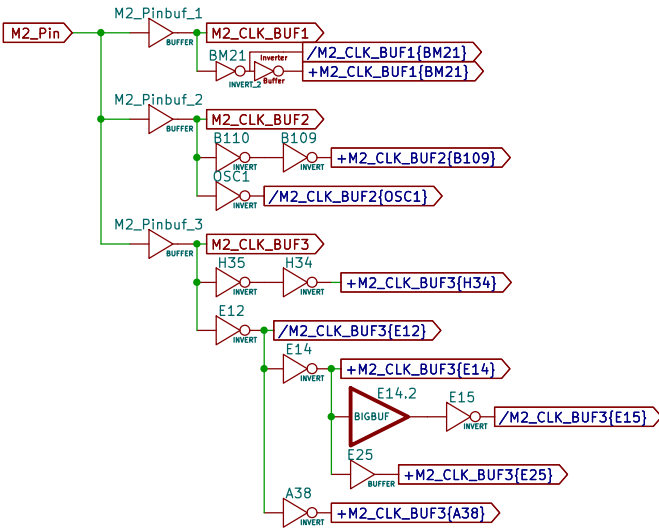
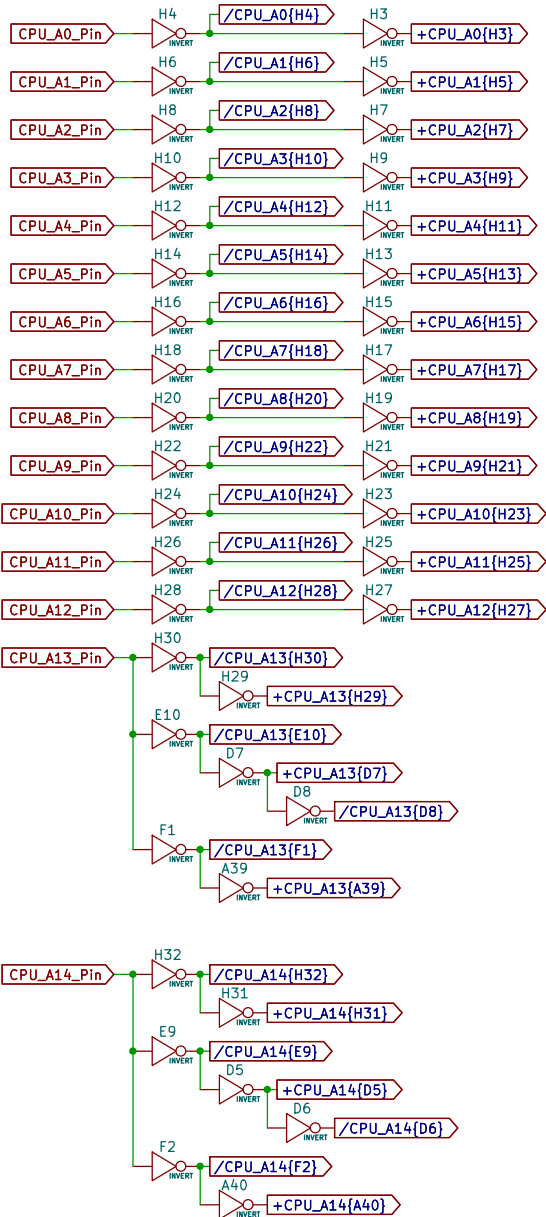
File: chr_address_bus_15_to_19.kicad_sch

Artifacts

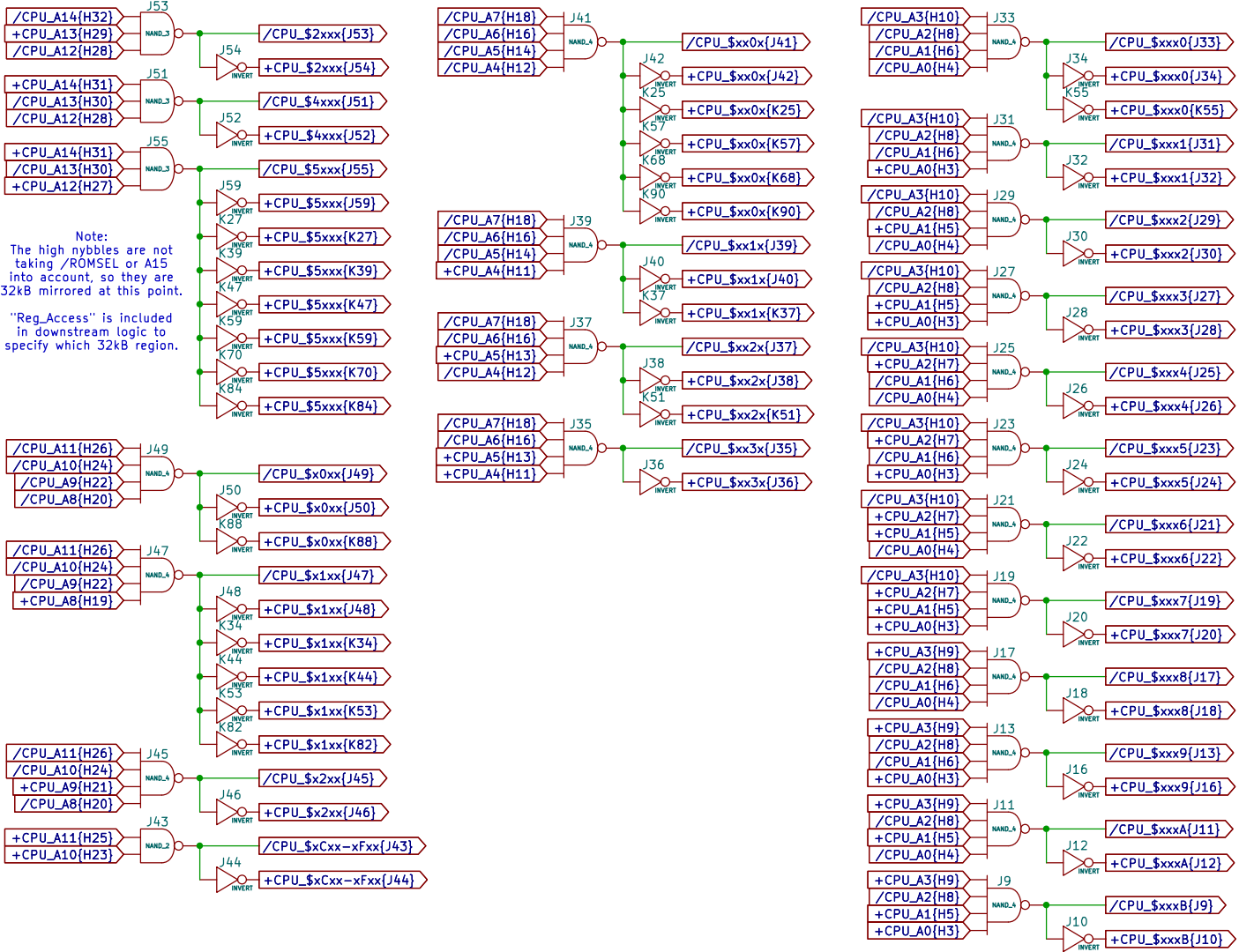
File: artifacts.kicad_sch



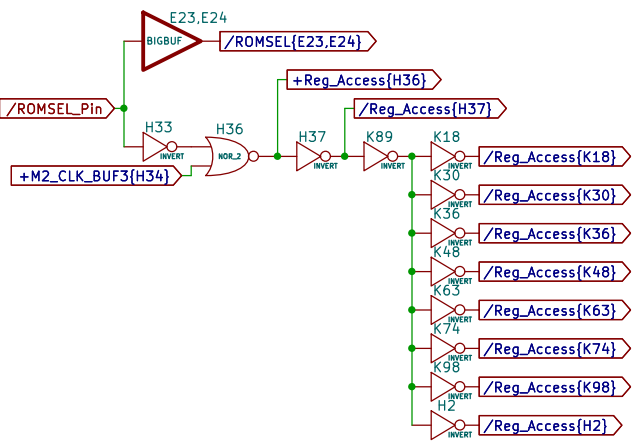
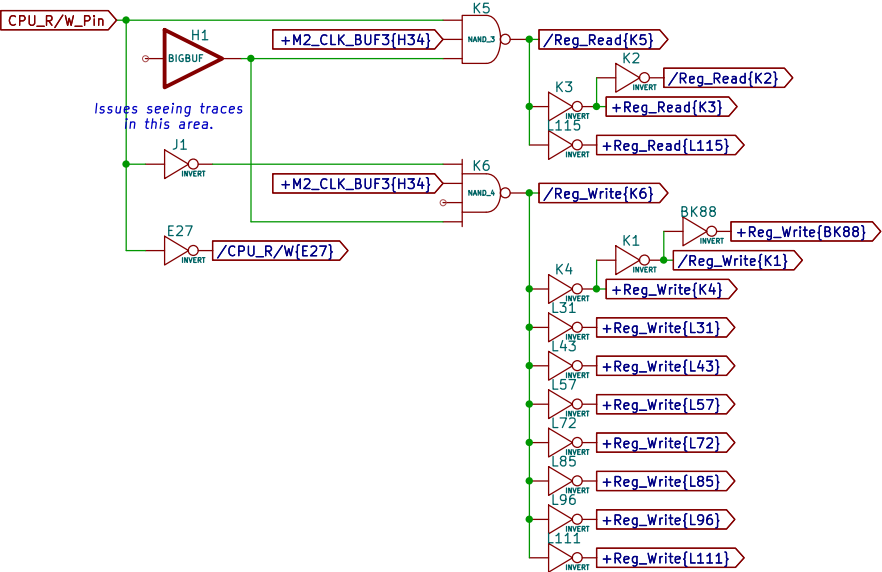
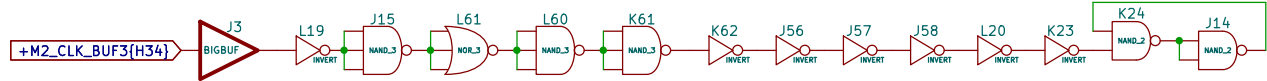
CPU Address Bus Buffering

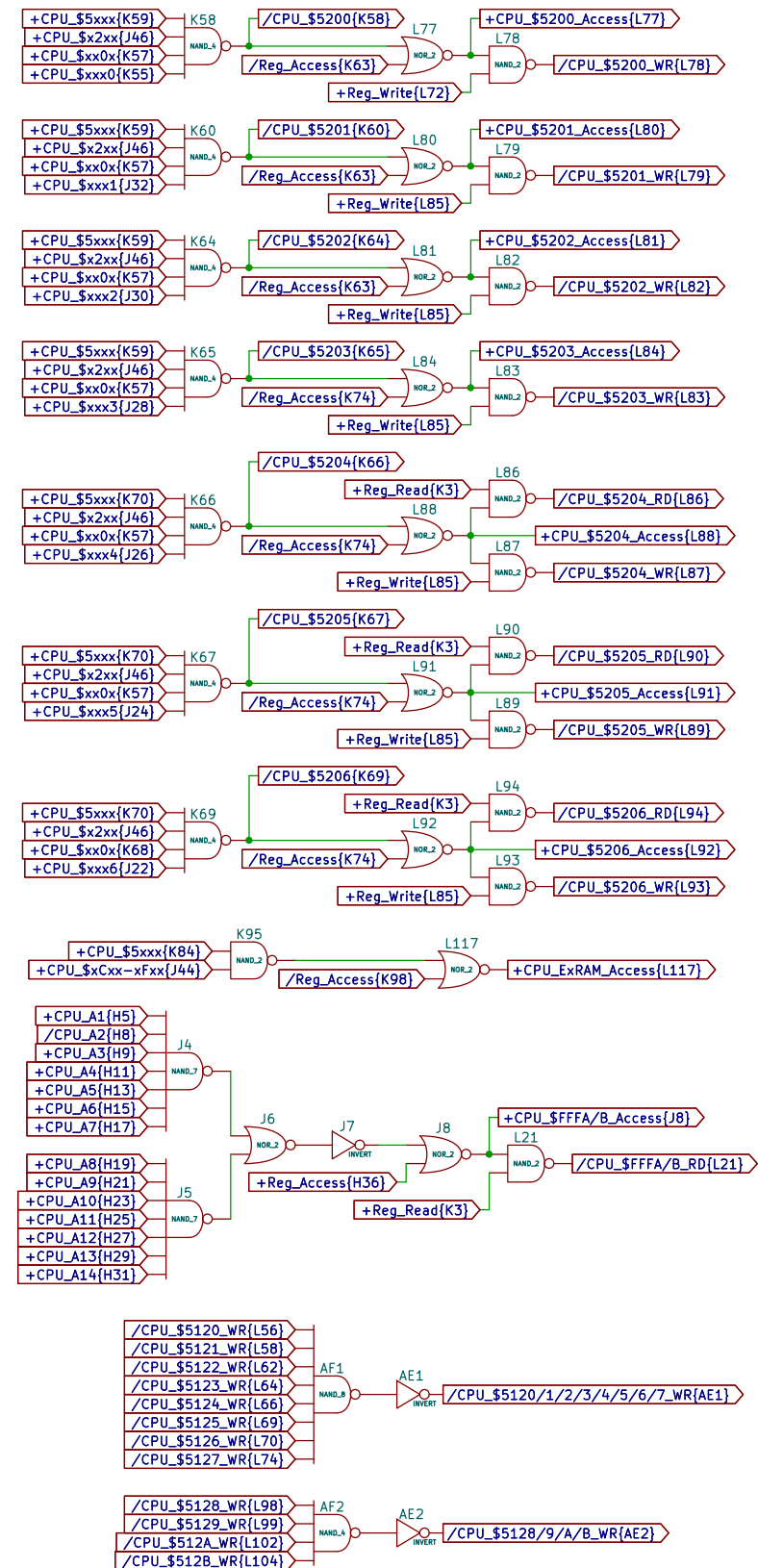
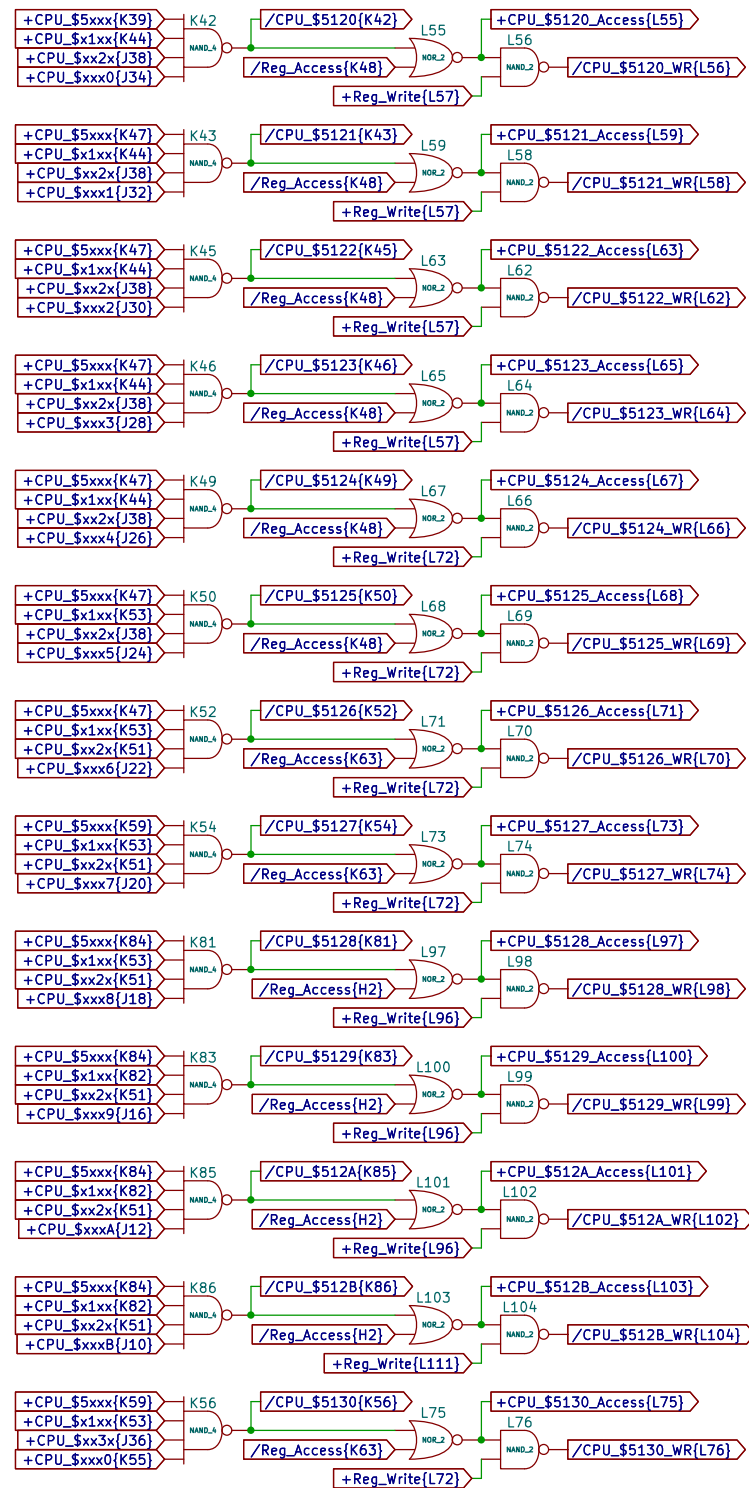
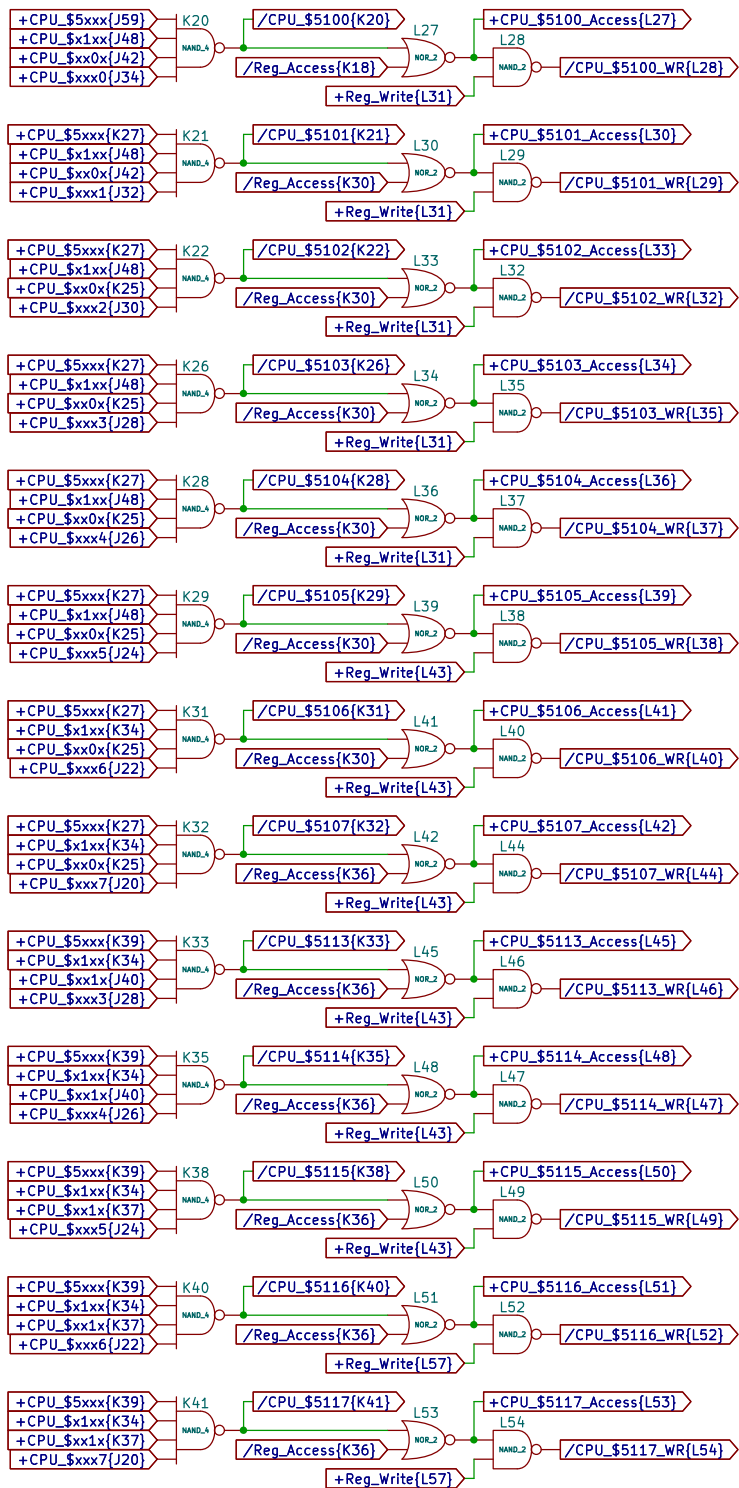
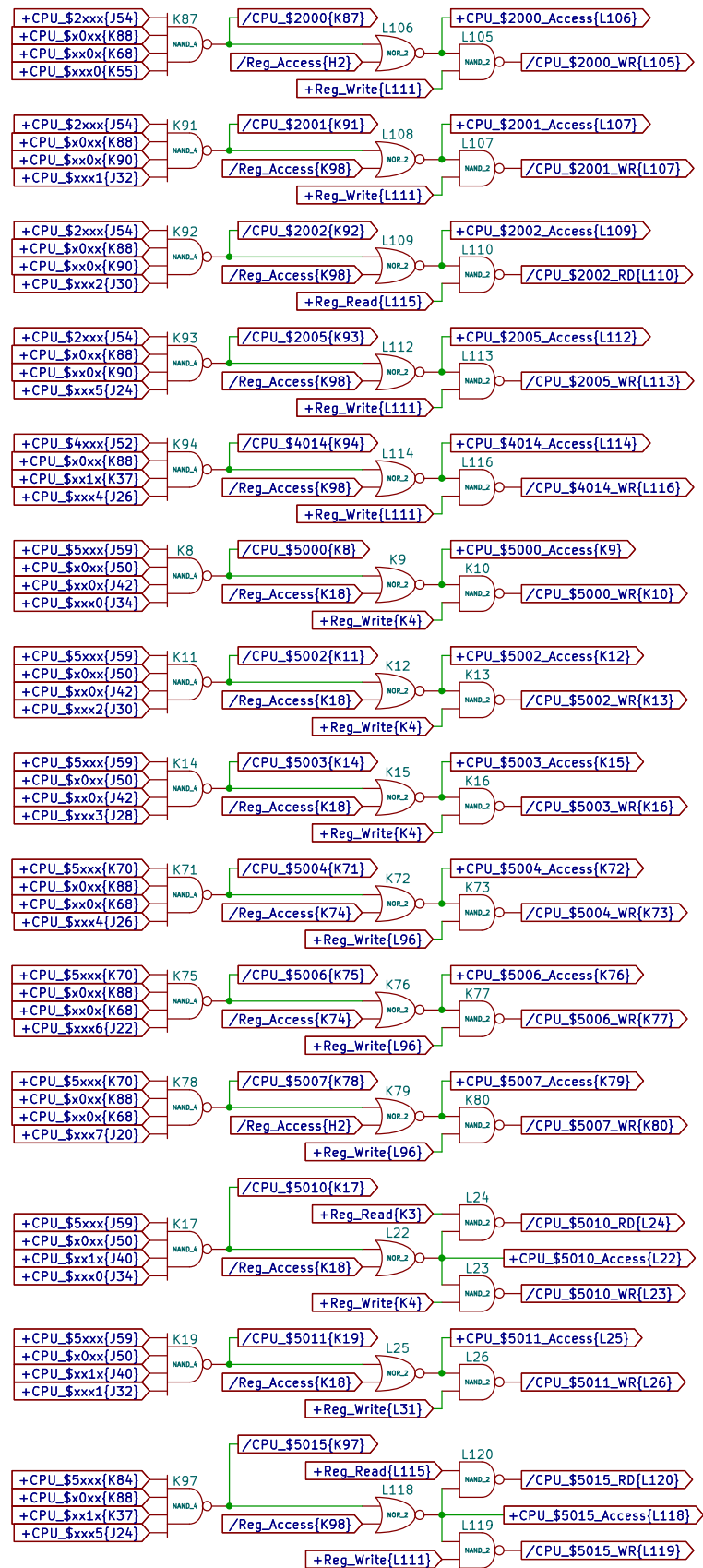


CPU Address Bus Nybble Decoding

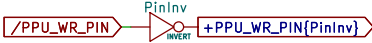
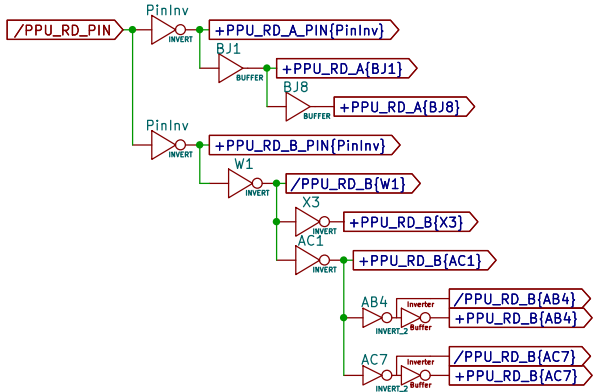
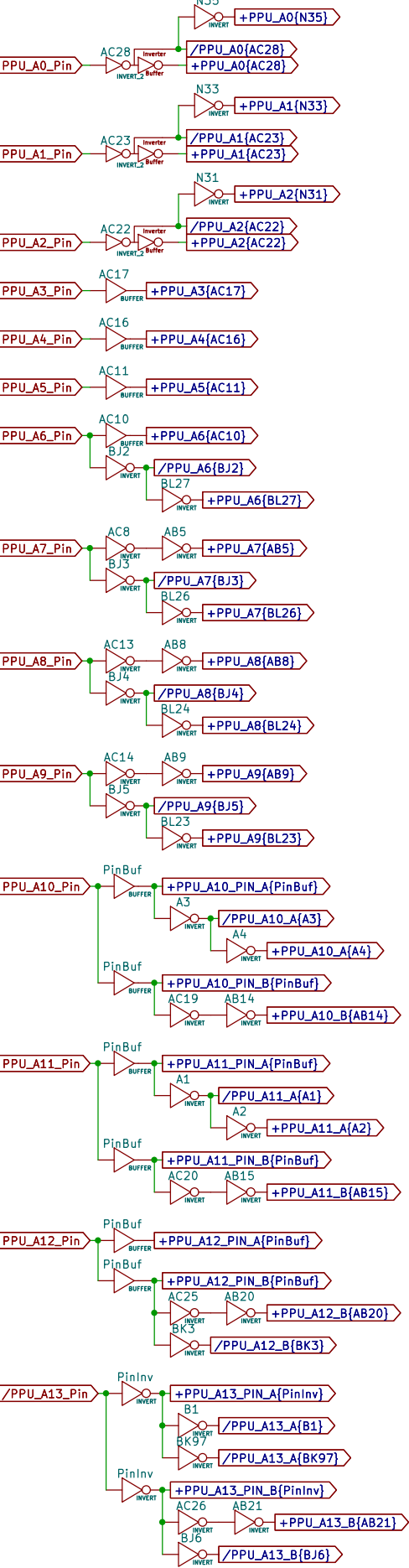


M2 Spaghetti That Goes Nowhere

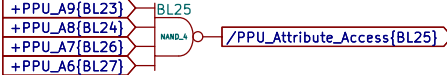


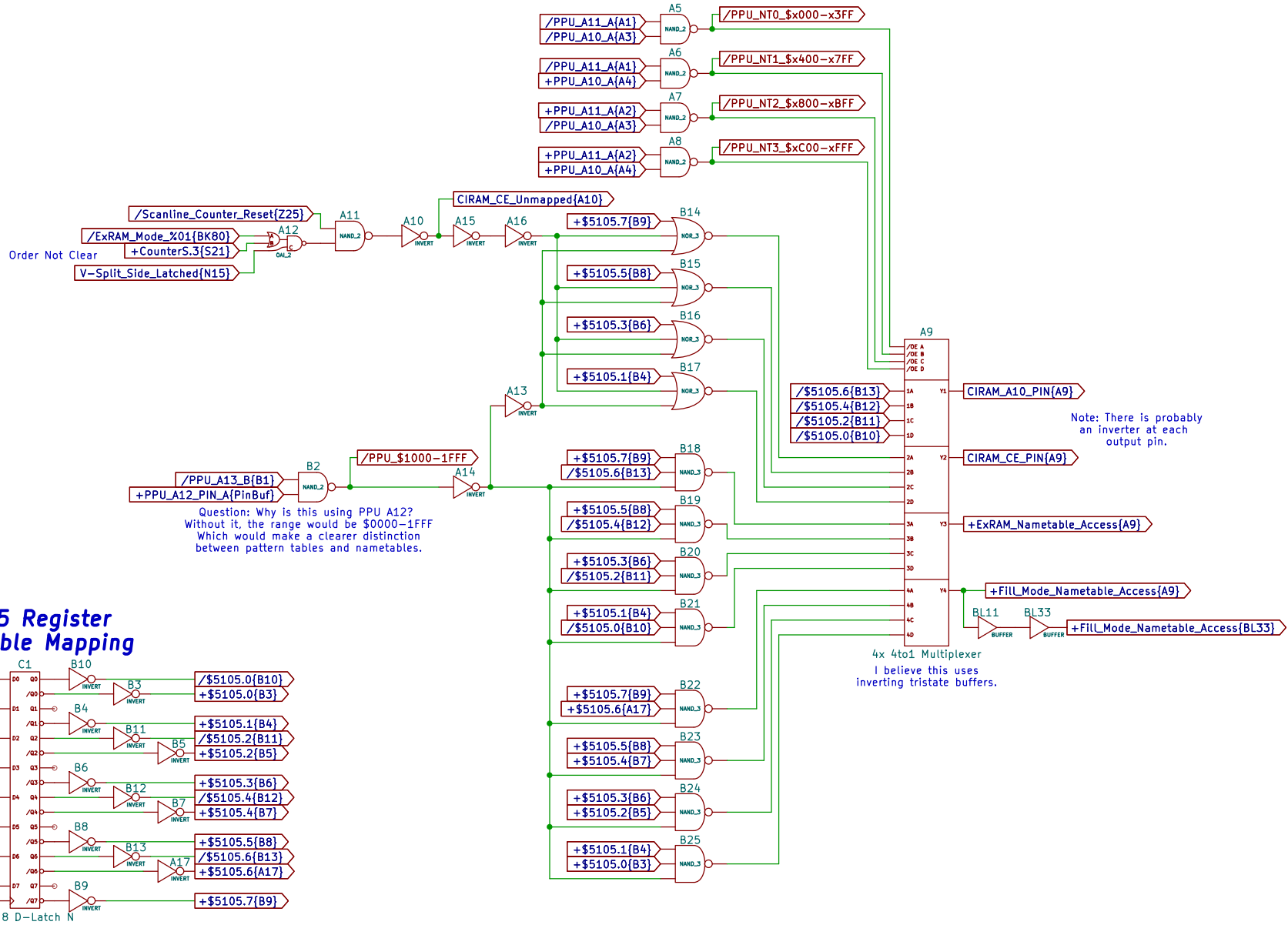


PPU Address Bus Buffering

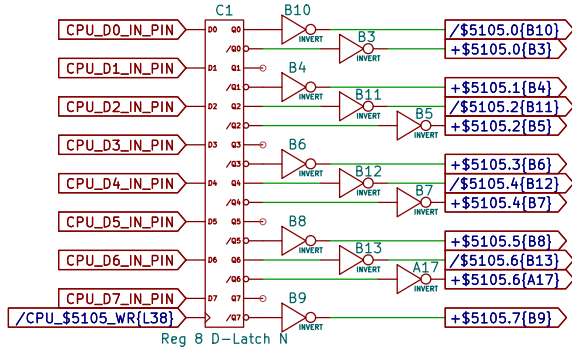


Note:
/PPU_WR_PIN has an inverter
at the pin but does not go
to any buffers beyond that.

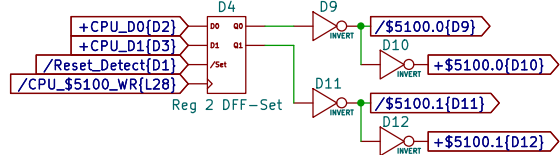




\$5105 Register Nametable Mapping

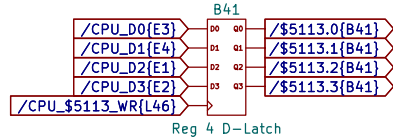


\$5100 Register PRG Mode

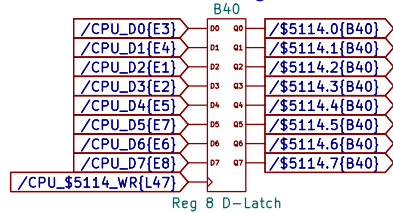


PRG Bank Registers

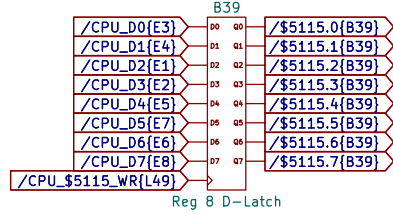
\$5113 Register



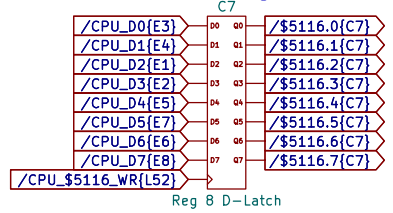
\$5114 Register



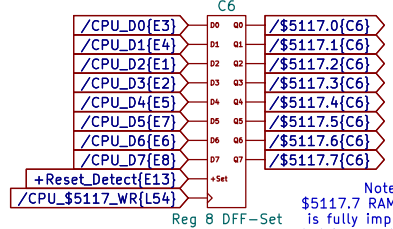
\$5115 Register



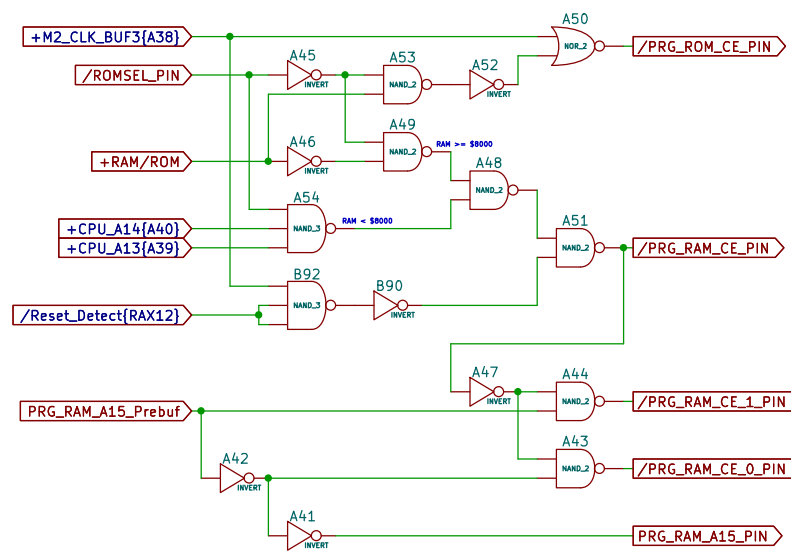
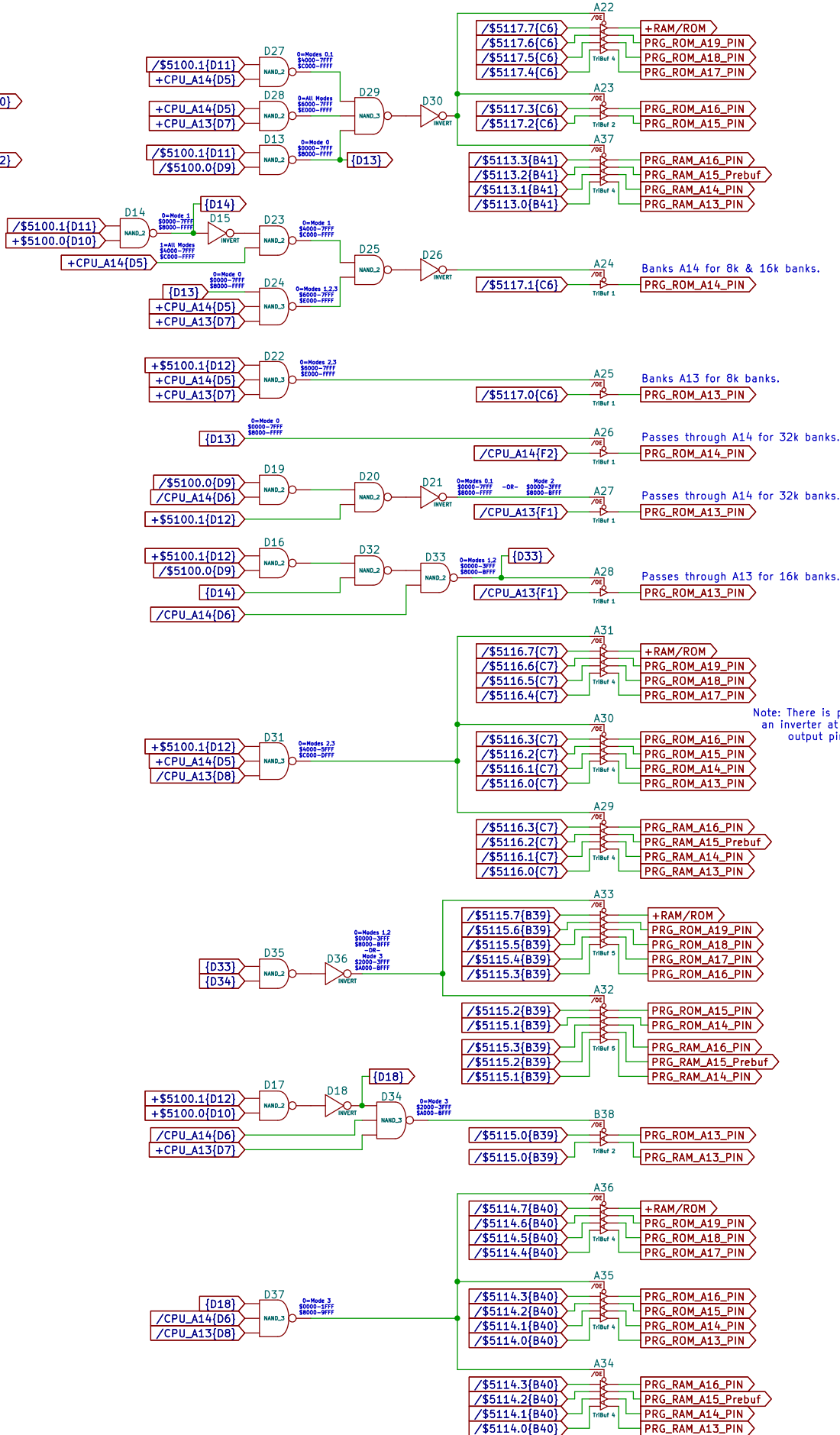
\$5116 Register



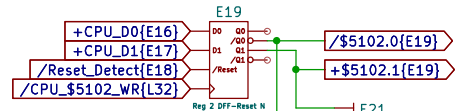
\$5117 Register



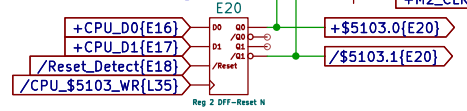
Note:
\$5117.7 RAM/ROM bit
is fully implemented
but is somehow being
prevented from entering
RAM mode when bench
testing.



\$5102 Register PRG-RAM Write Protection

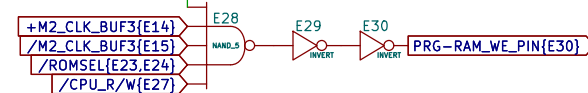


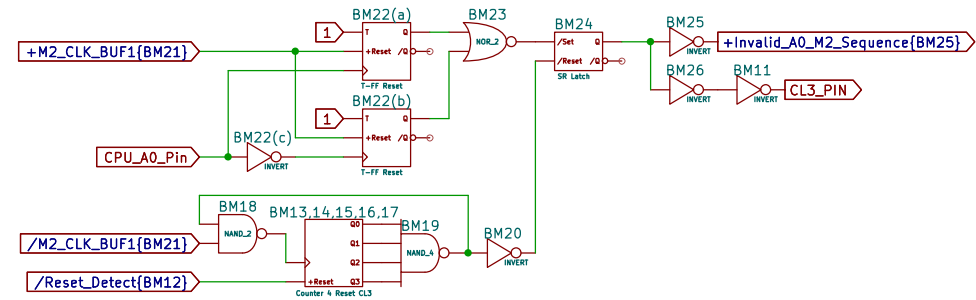
\$5103 Register



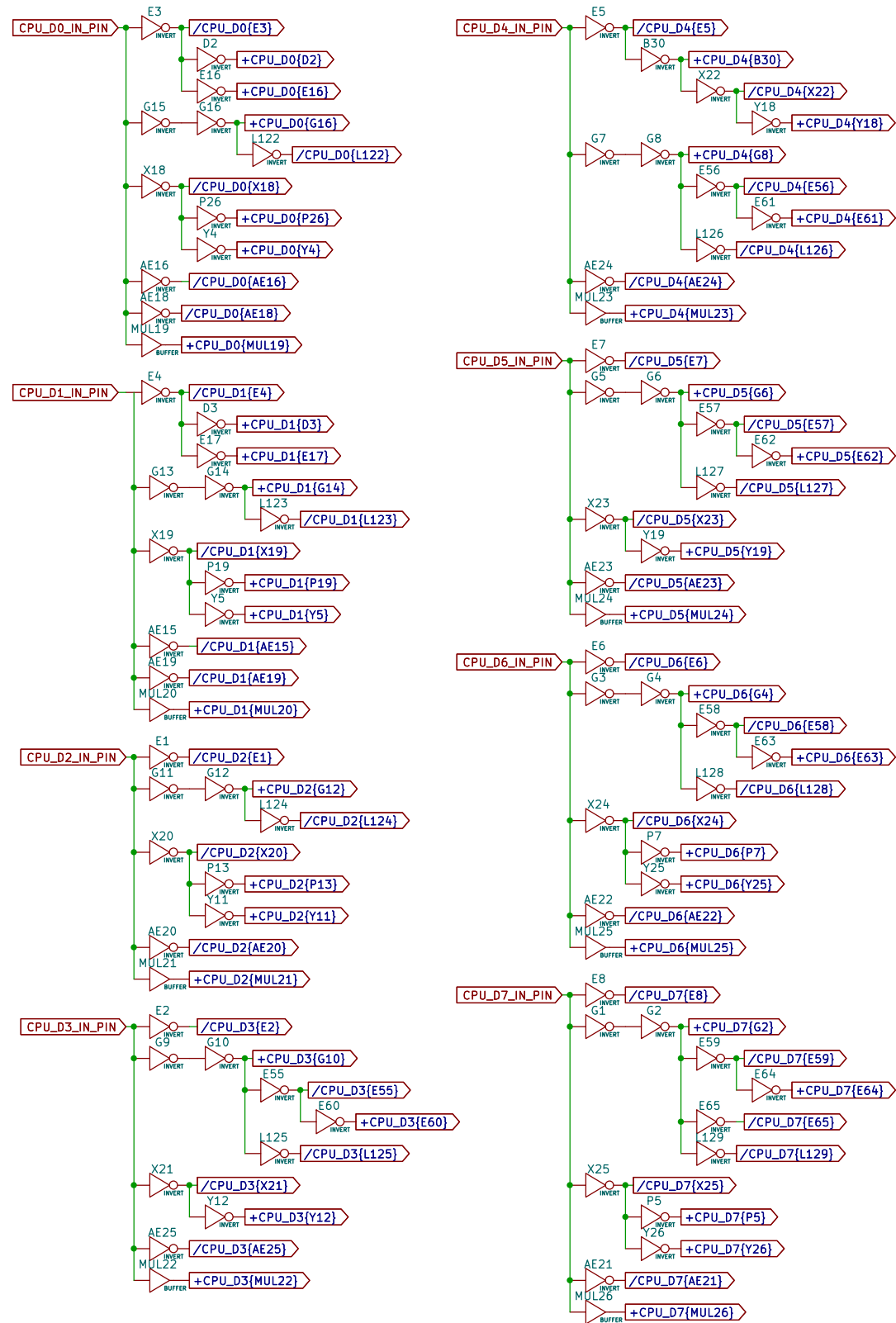
Note: /Set vs. /Reset inputs are individually used
but not easily shown here with a symbol.

Unlocked when
E22=1
Q becomes locked asynchronously,
becomes unlocked synchronized to M2

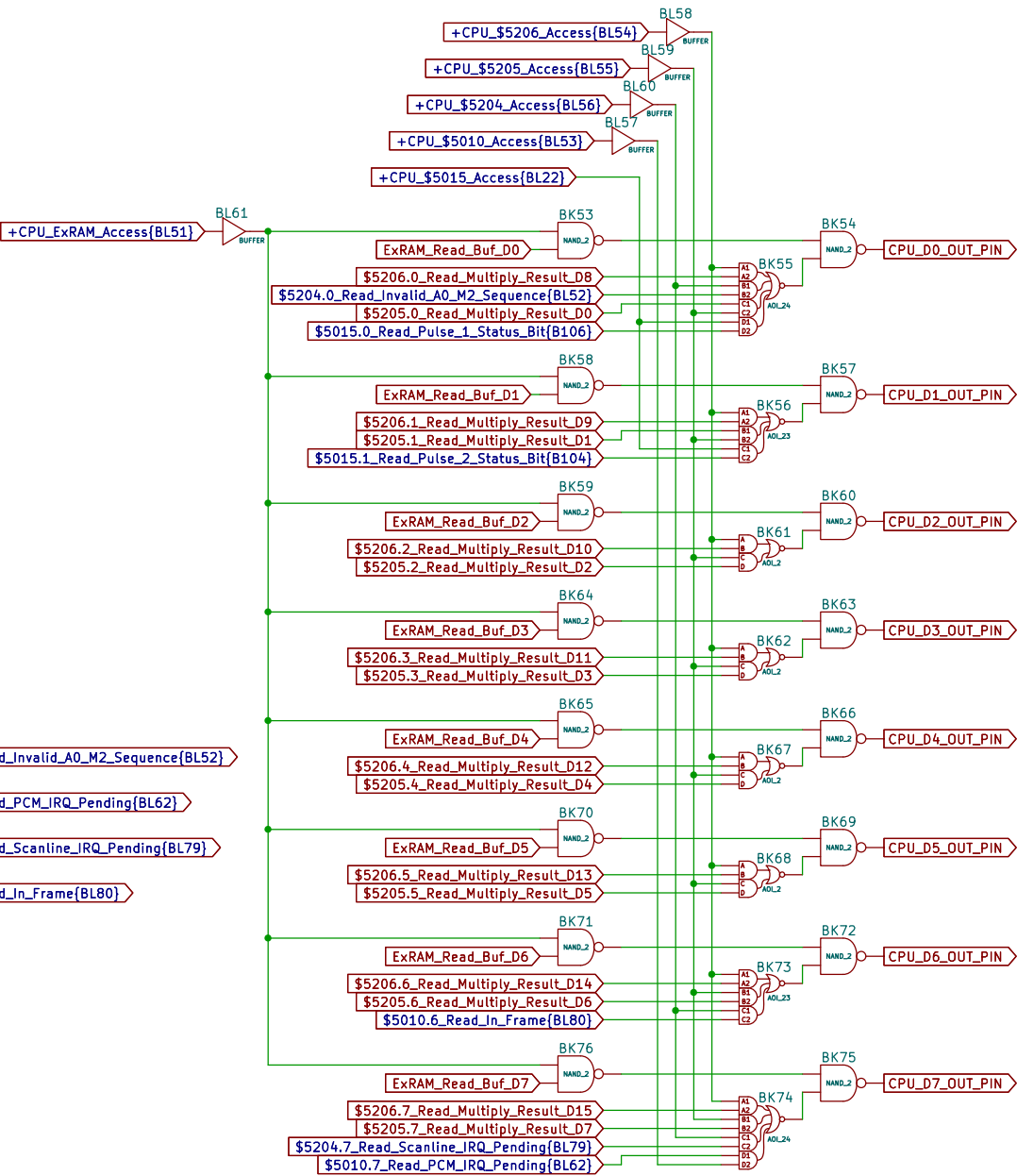




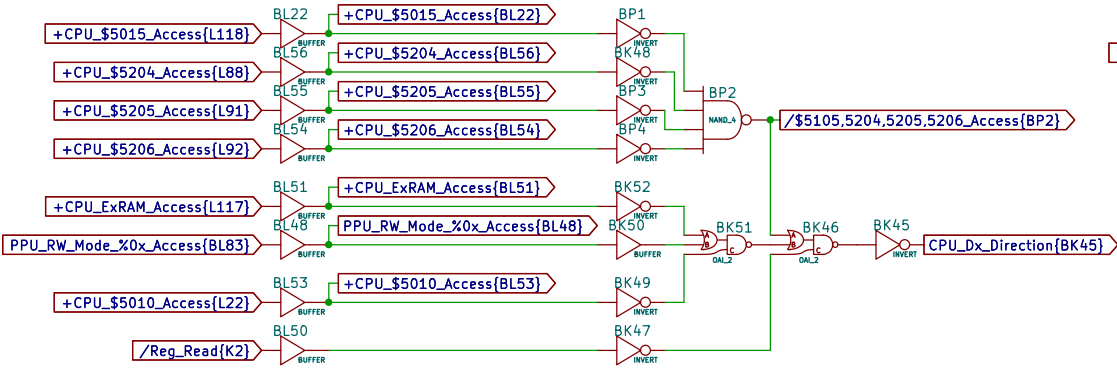
Incoming CPU Data Bus Buffering Bonanza



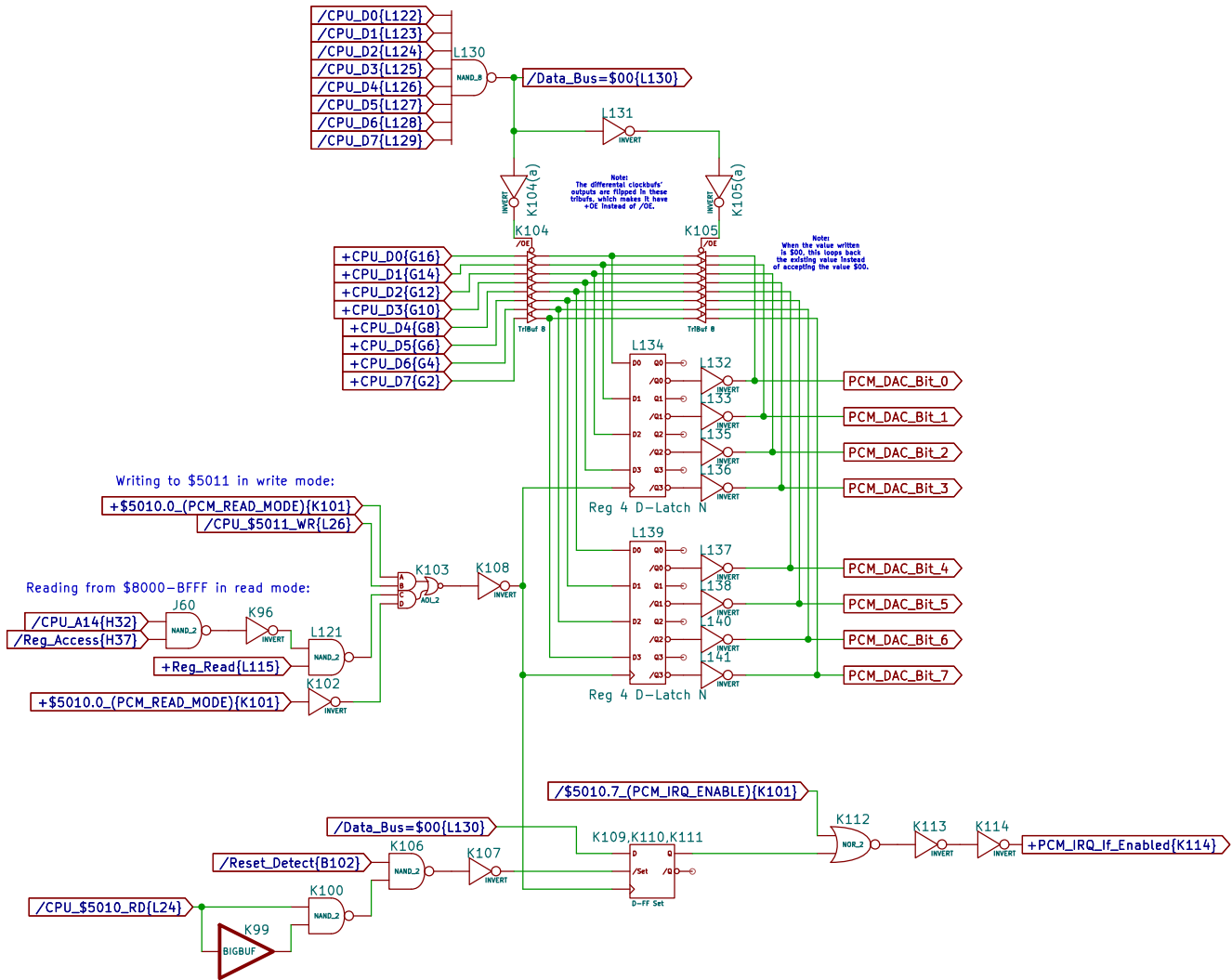
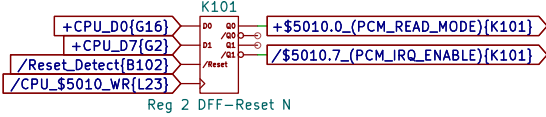
Multiplexing Data to be Driven onto the CPU Data Bus



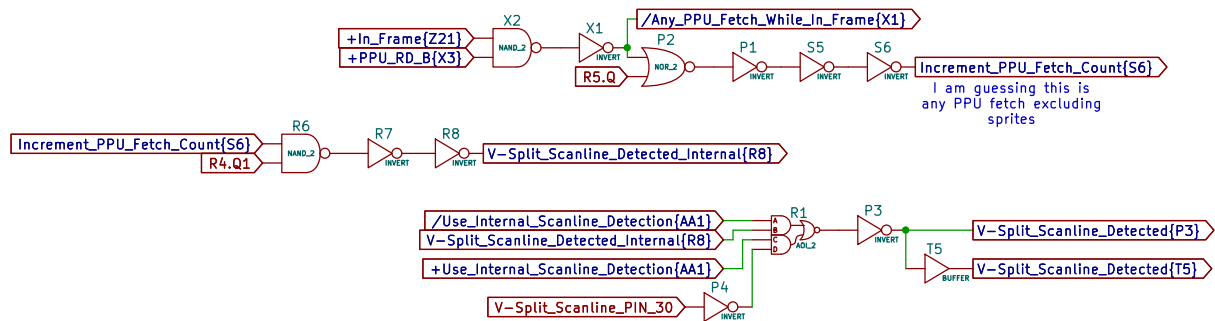
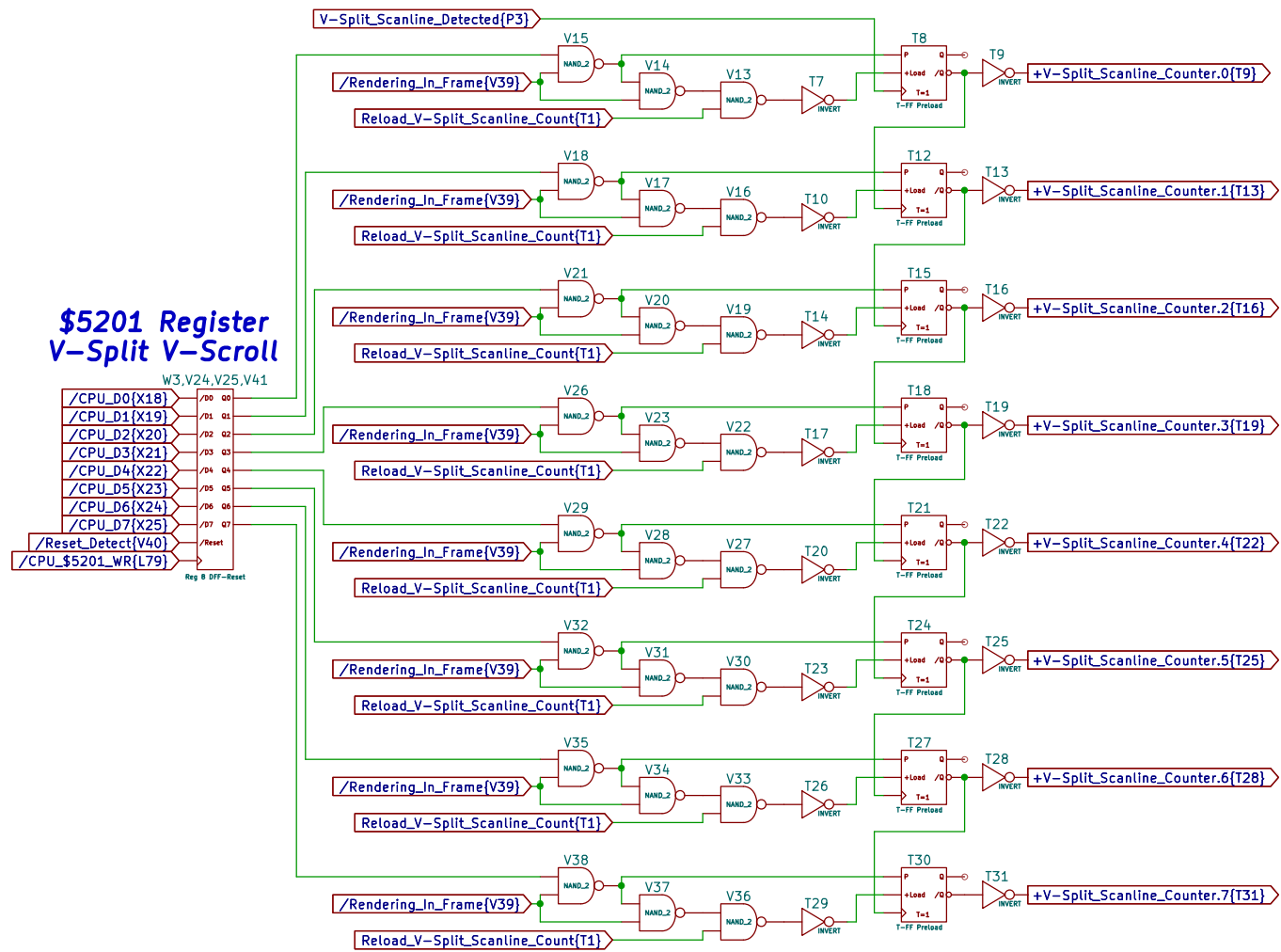
Decides When to Drive the CPU Data Bus



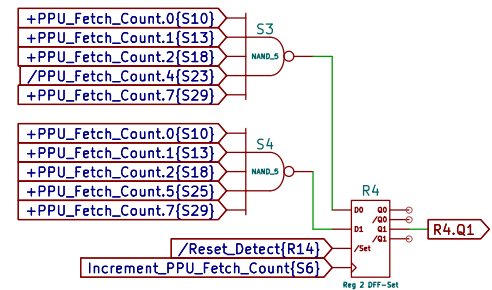
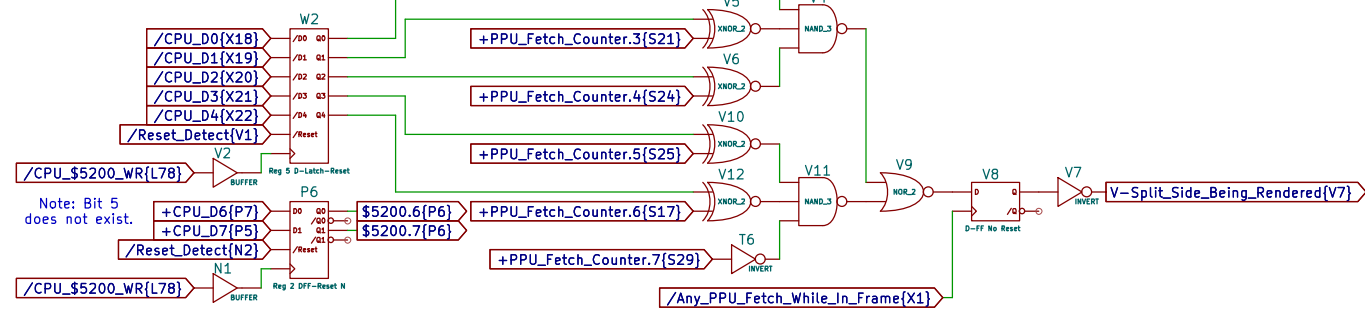
\$5010 Register
Mode & IRQ Enable



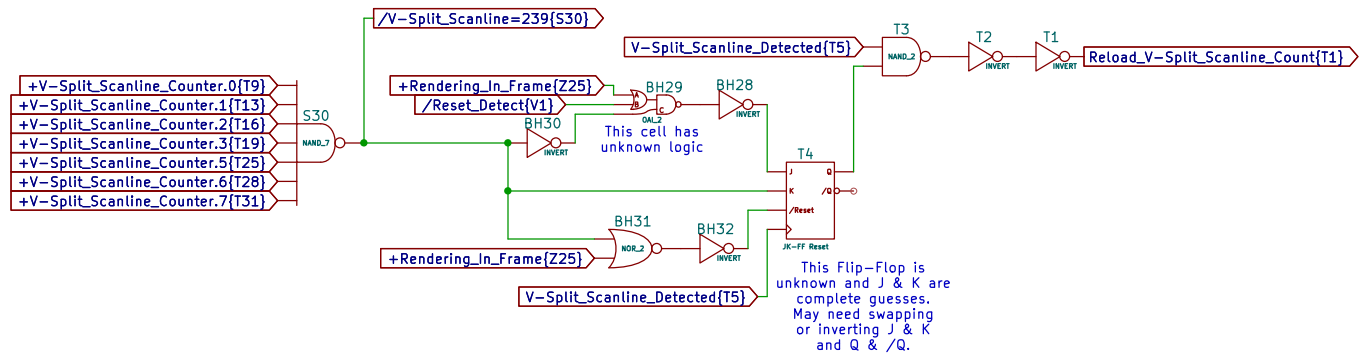
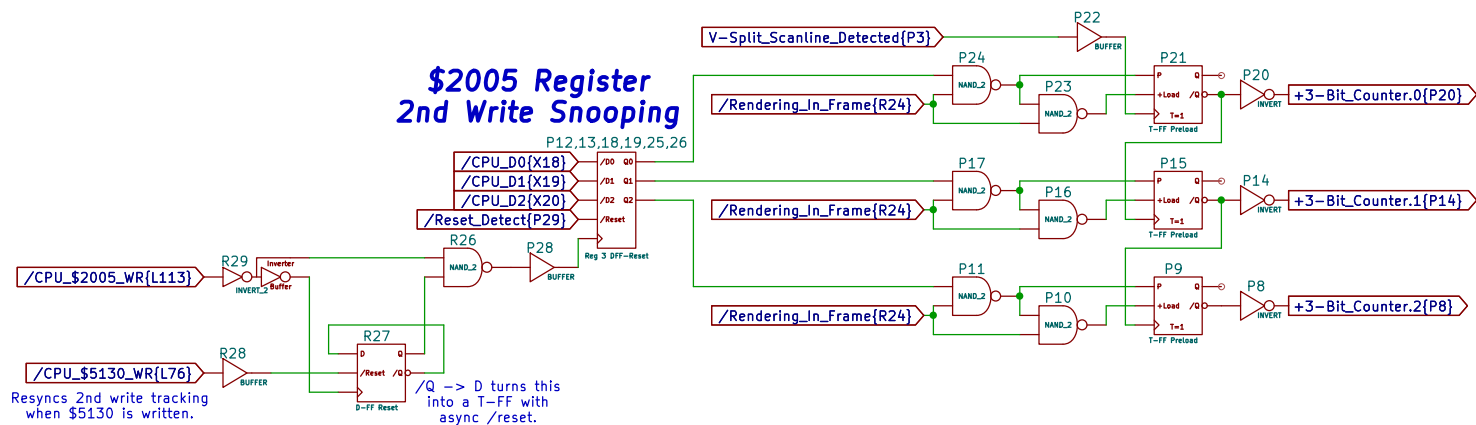
V-Split Scanline Counter



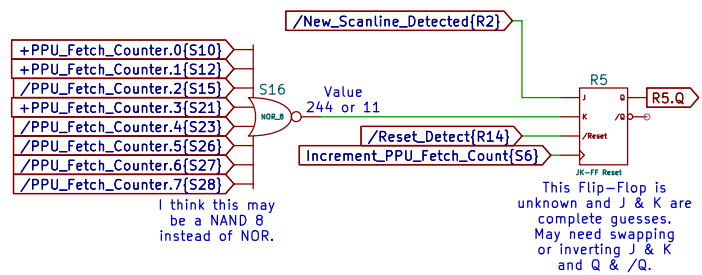
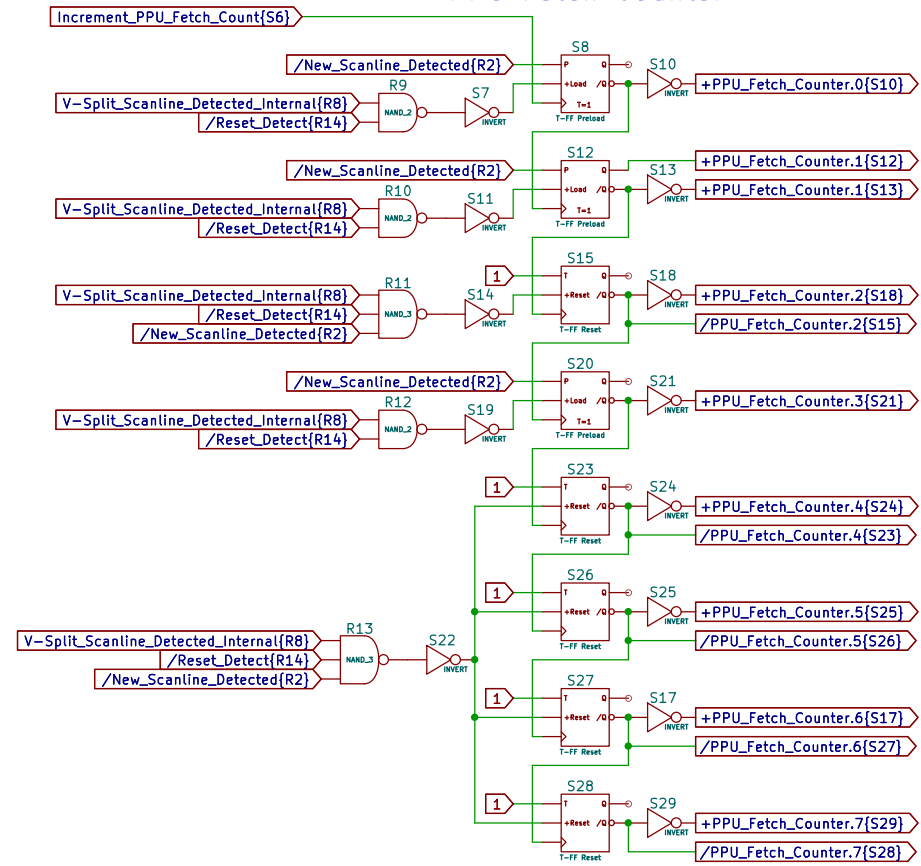
\$5200 Register V-Split Mode & H-Threshold

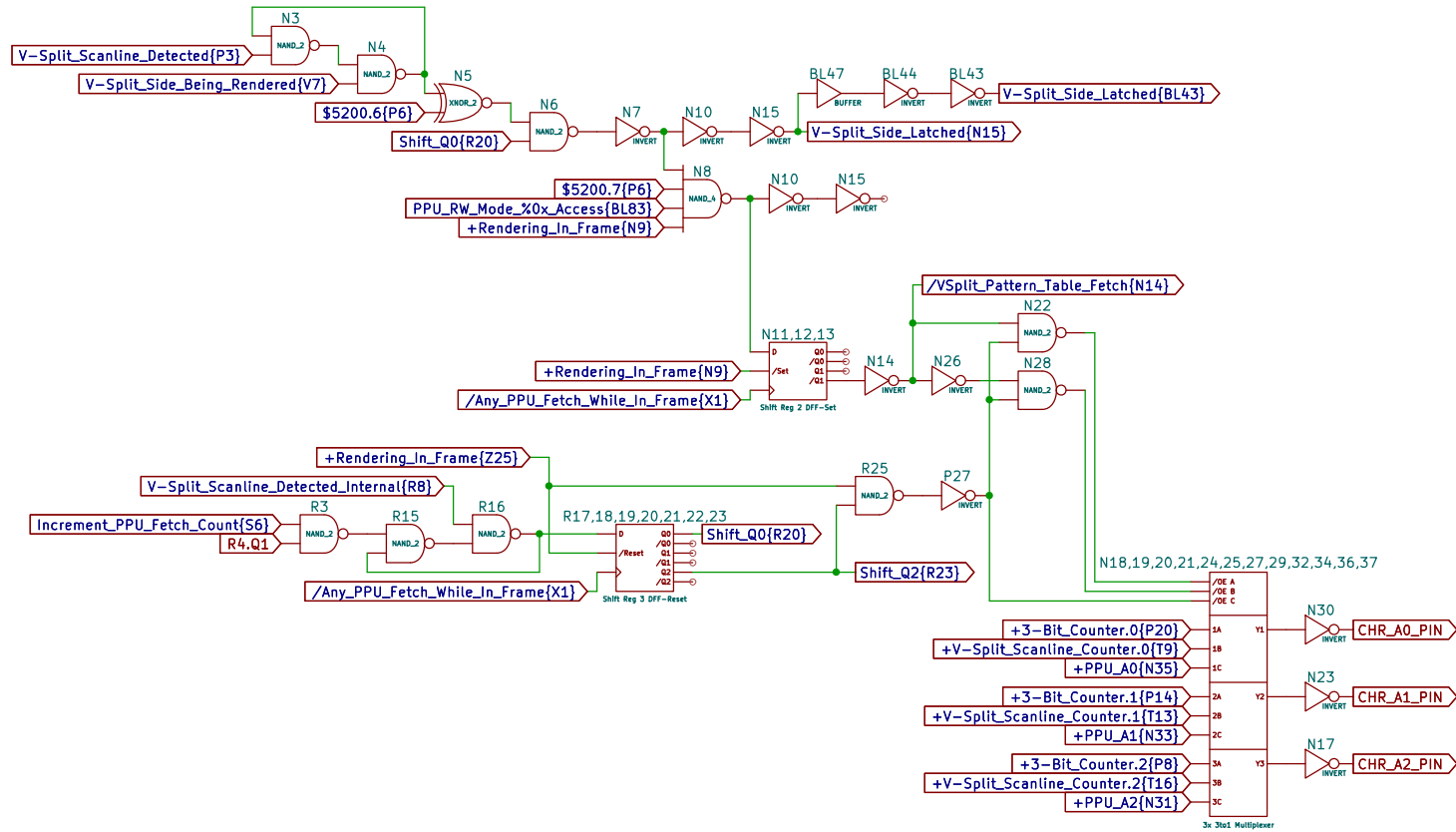


Unknown 3-Bit Scanline Counter

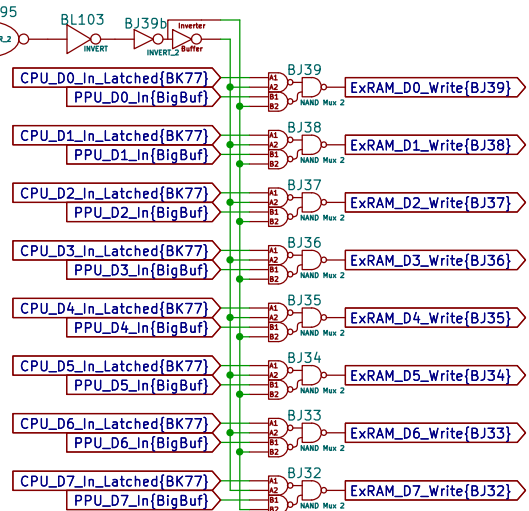
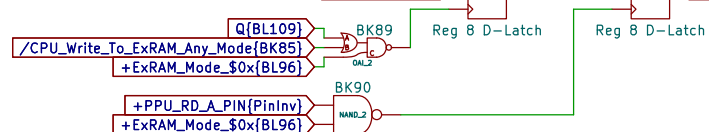
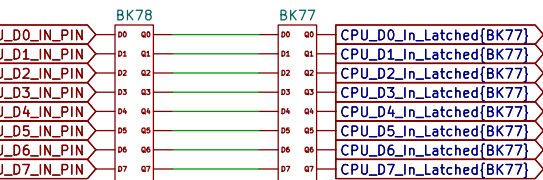
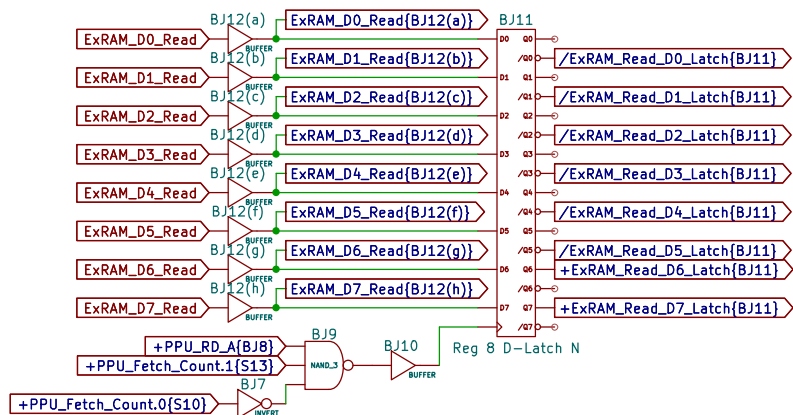
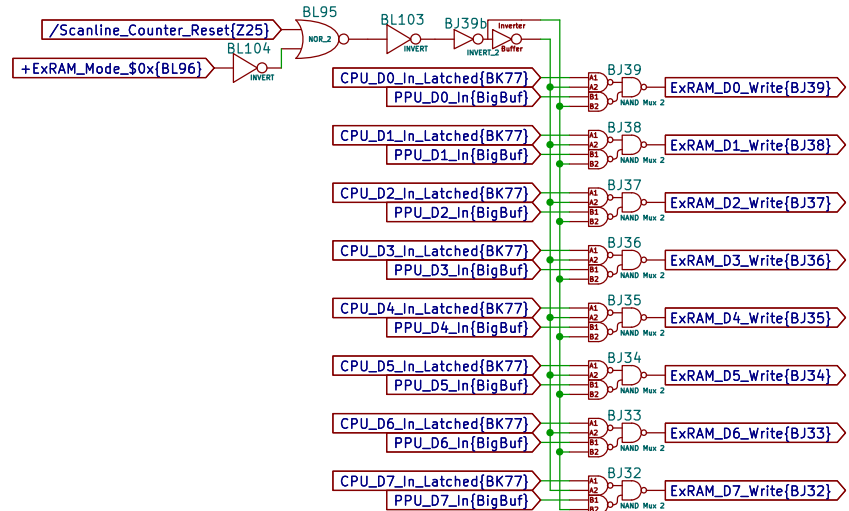
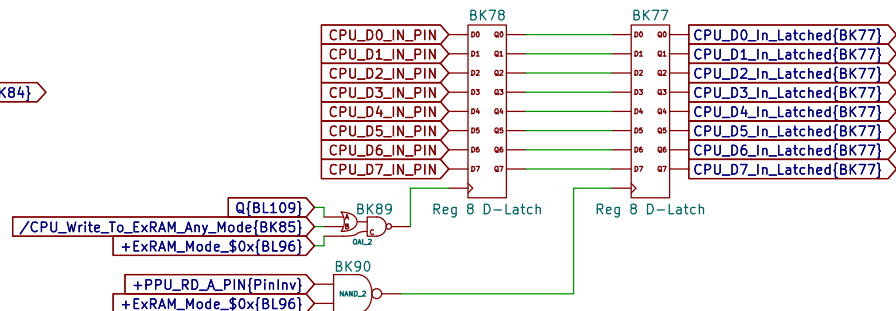
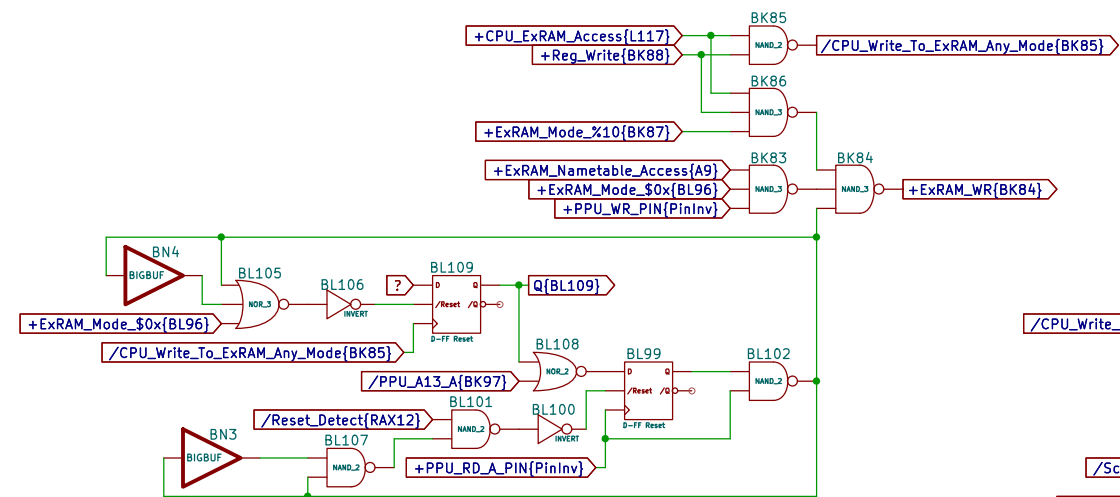
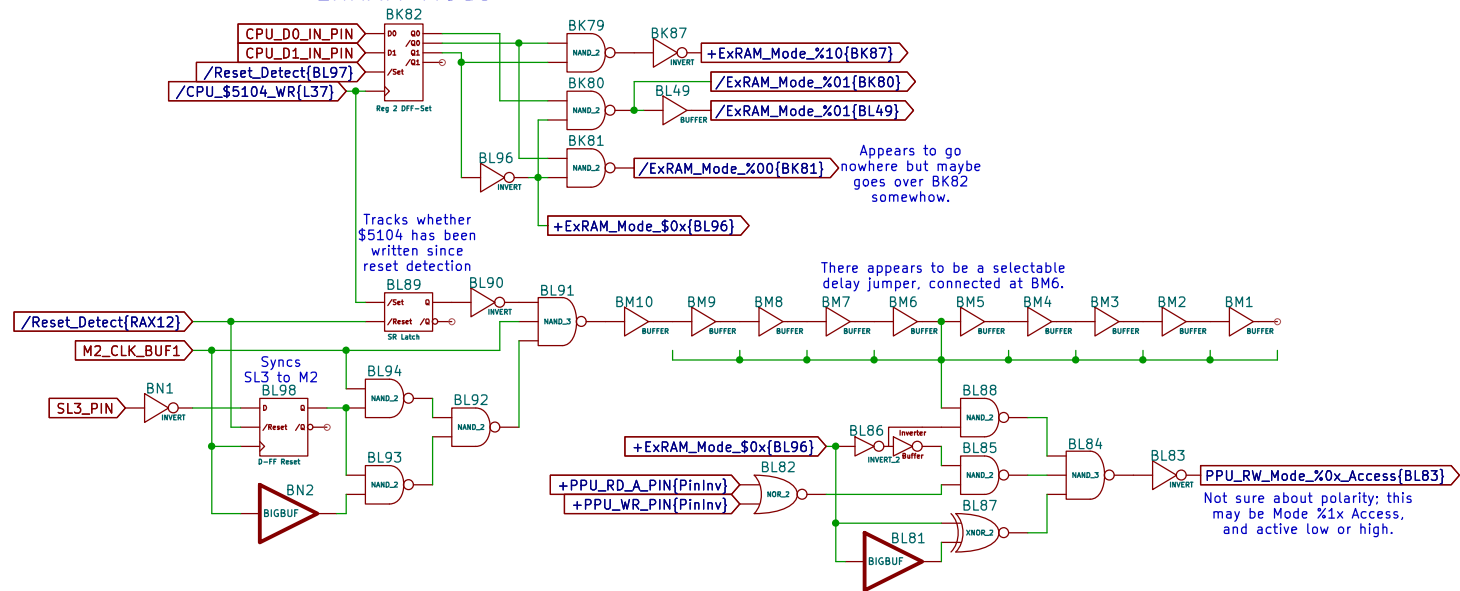


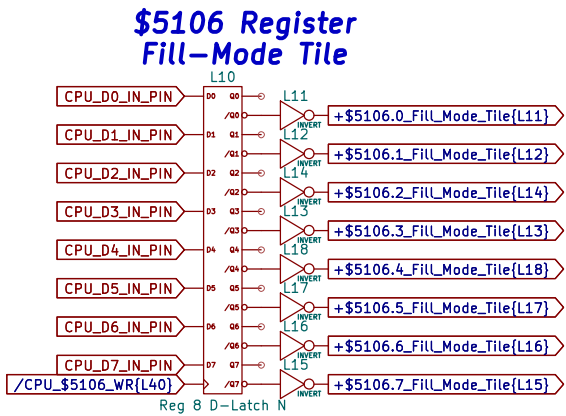
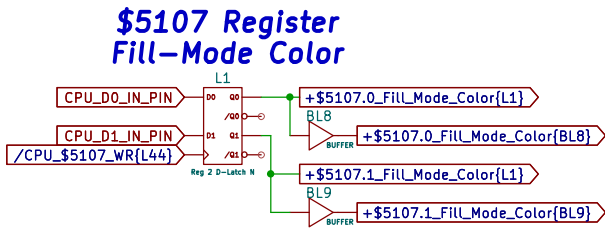
PPU Fetch Counter



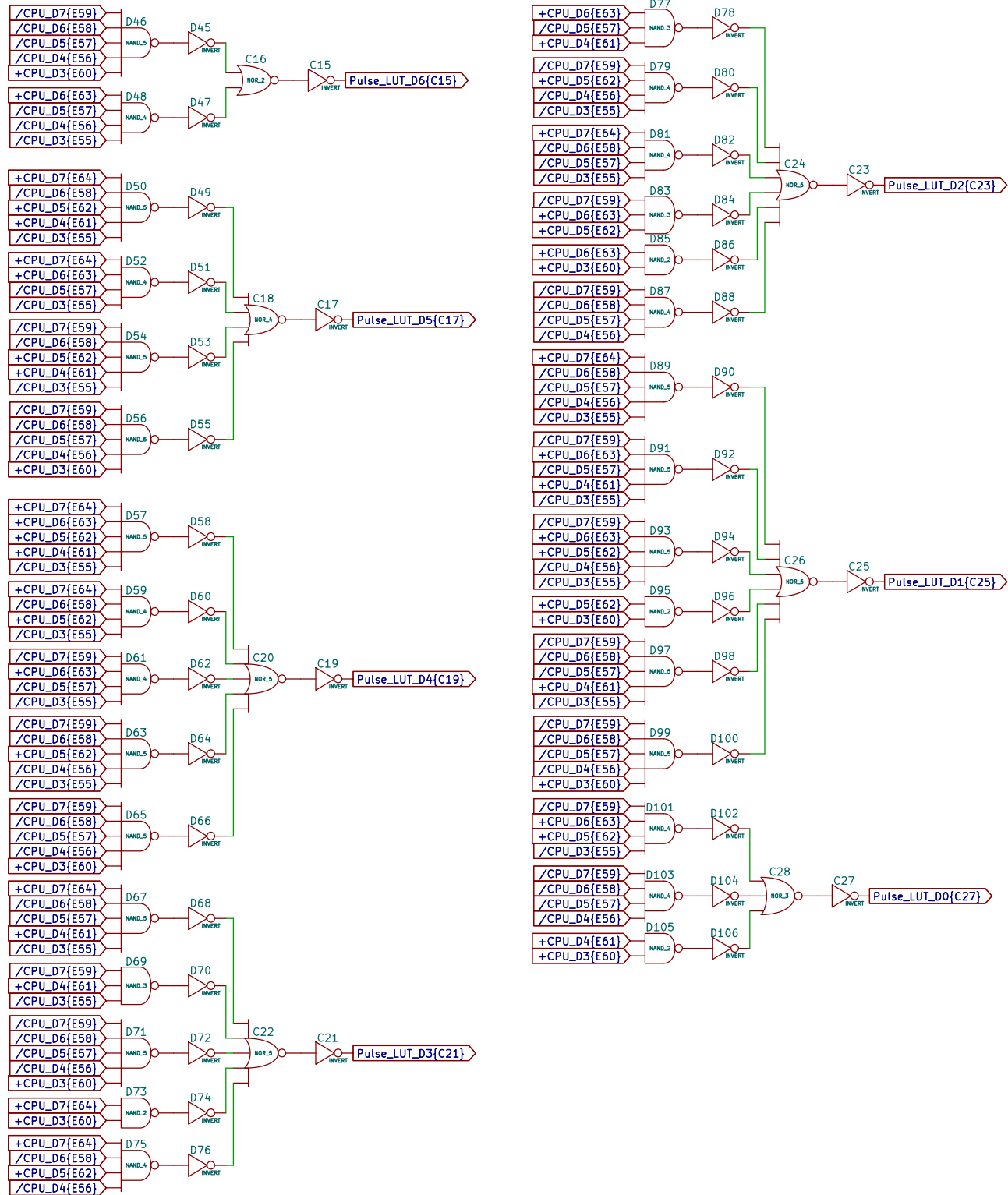


\$5104 Register ExRAM Mode

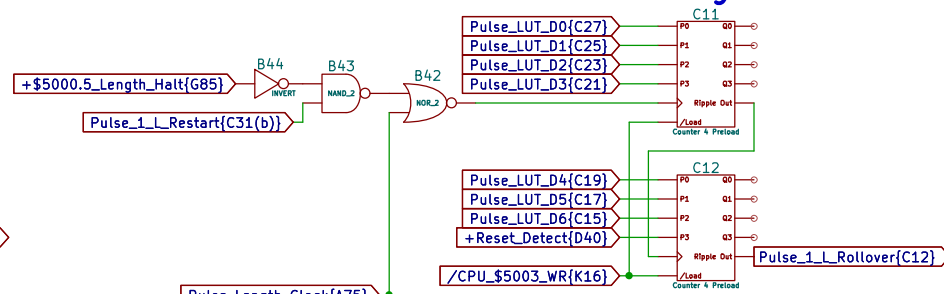




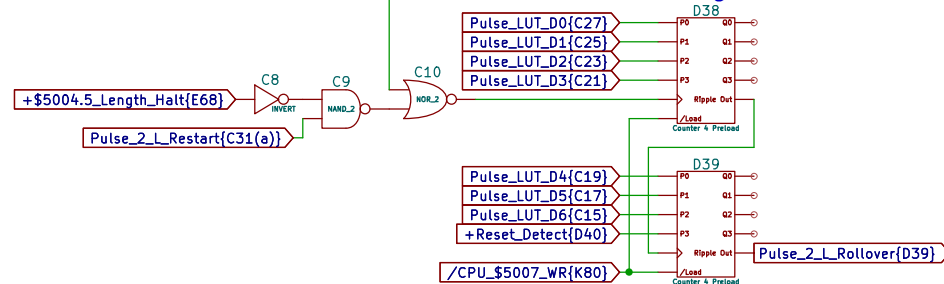
5-Bit to 7-Bit Pulse Length Lookup Table



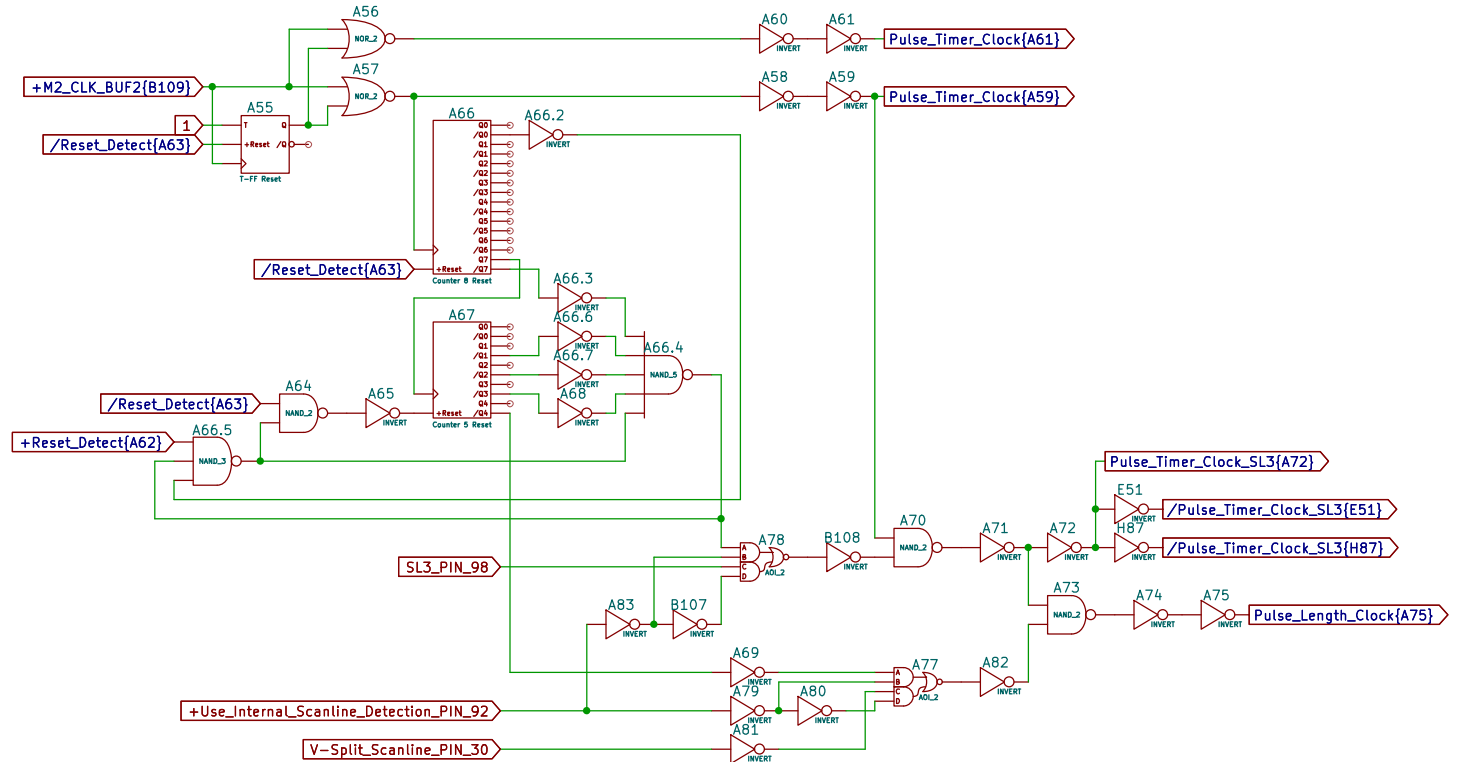
\$5003 Register Pulse 1 Length Counter



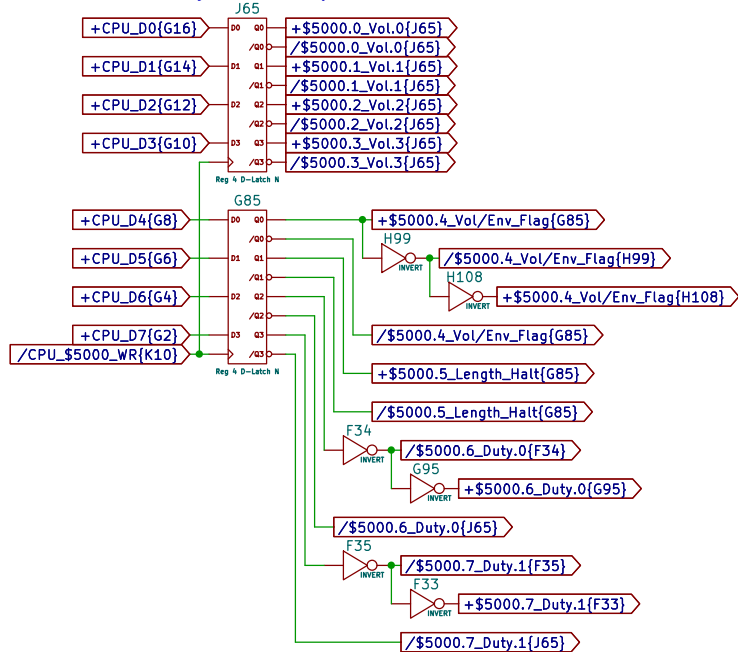
\$5007 Register Pulse 2 Length Counter



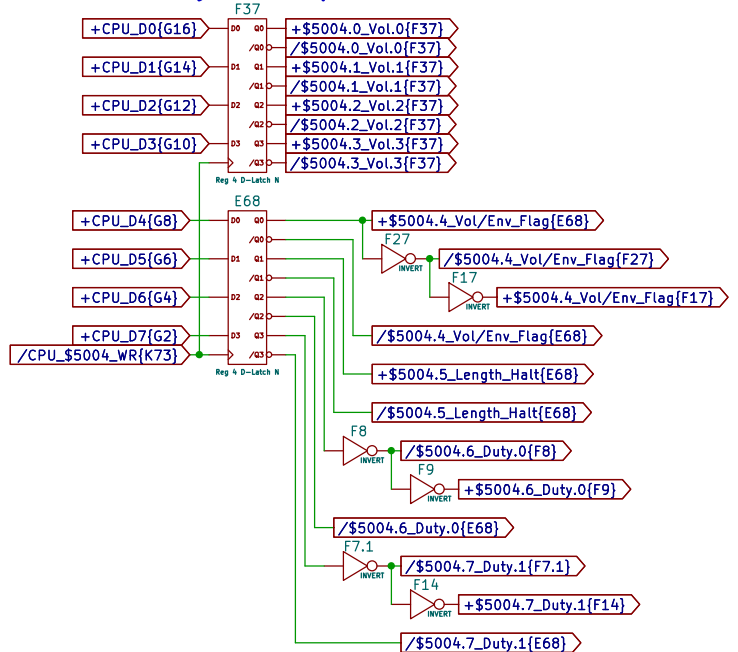
Pulse Channel Clock Source



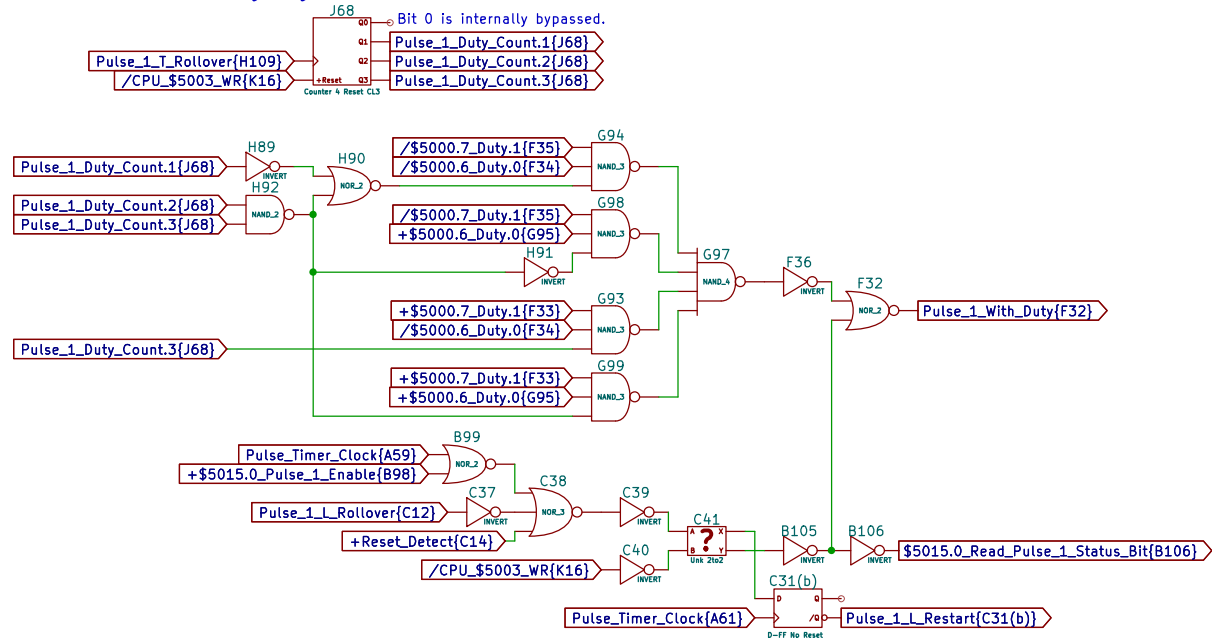
\$5000 Register
Pulse 1 Duty Envelope Volume



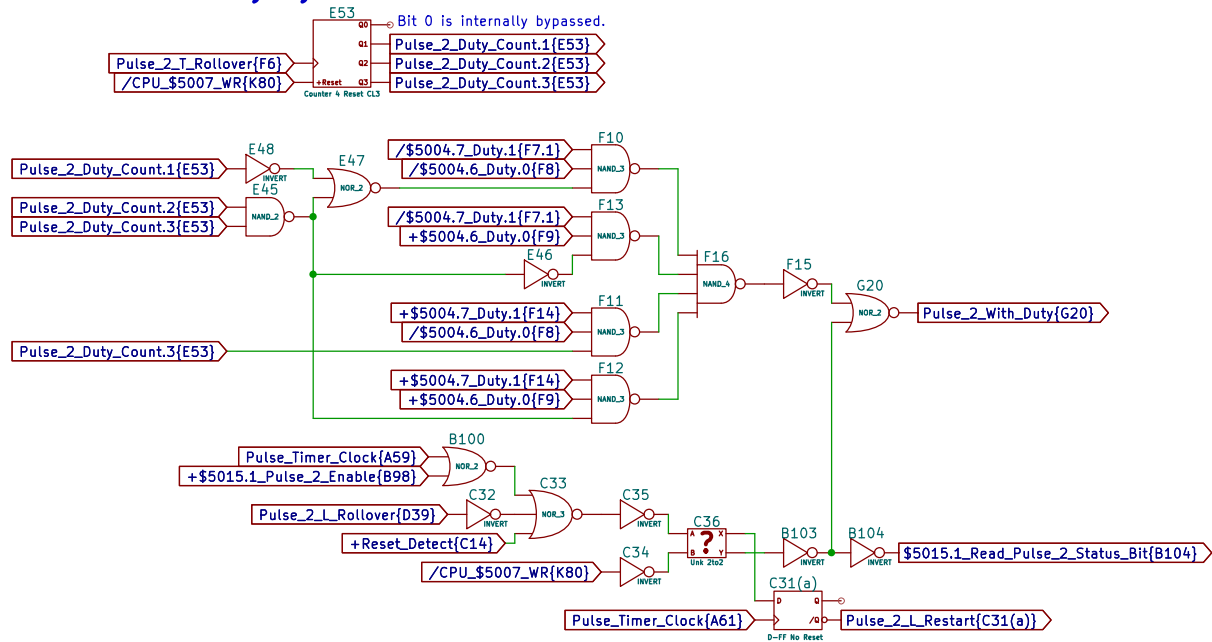
\$5004 Register
Pulse 2 Duty Envelope Volume



Pulse 1 Duty Cycle Counter

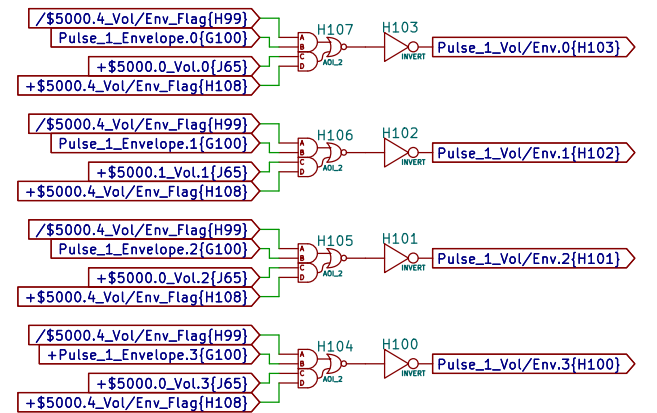


Pulse 2 Duty Cycle Counter



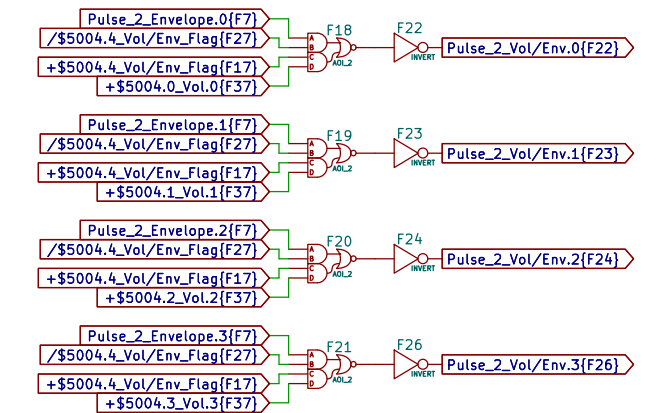
Pulse 1

Envelope/Volume Multiplexing

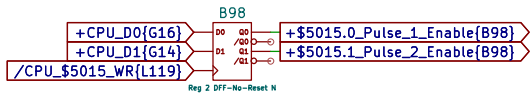


Pulse 2

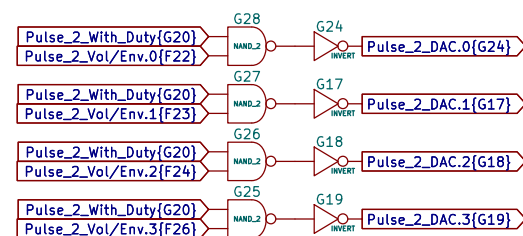
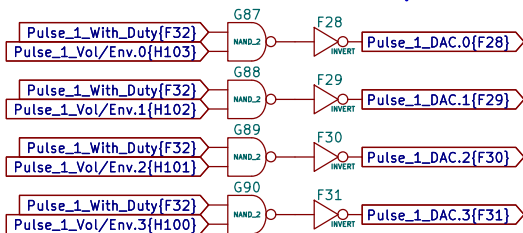
Envelope/Volume Multiplexing



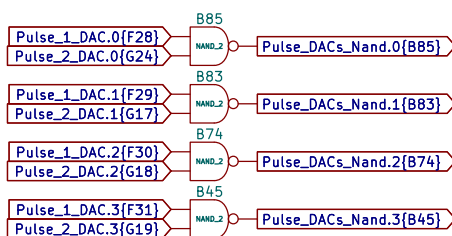
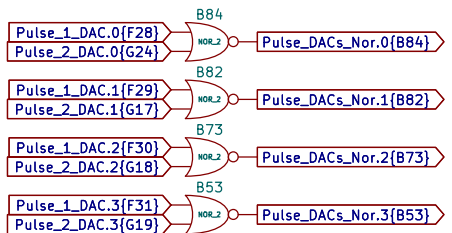
\$5015 Write Register Pulse 1&2 Status



Pulse 1&2 Individual 4-bit DAC Outputs

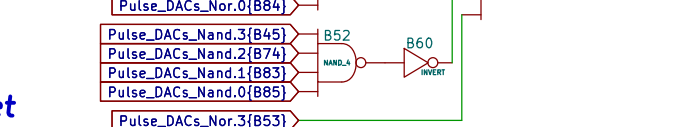
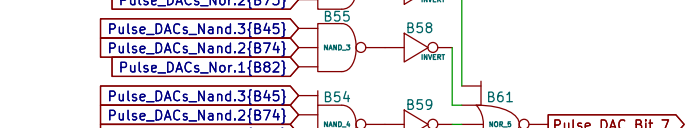
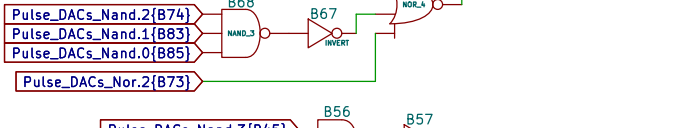
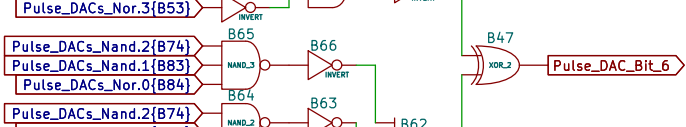
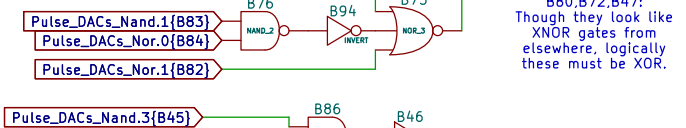
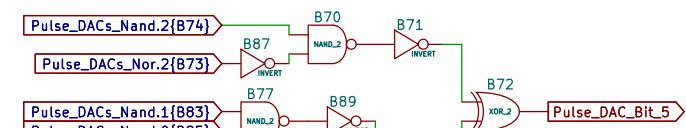
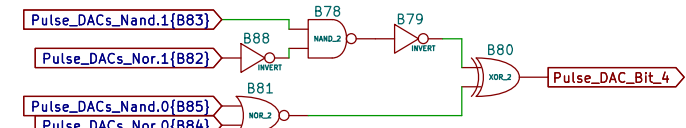
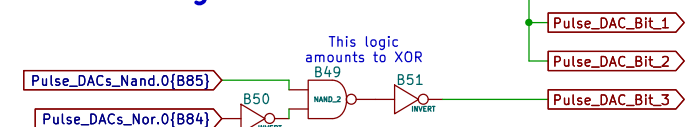


Components for 4-bit Adder

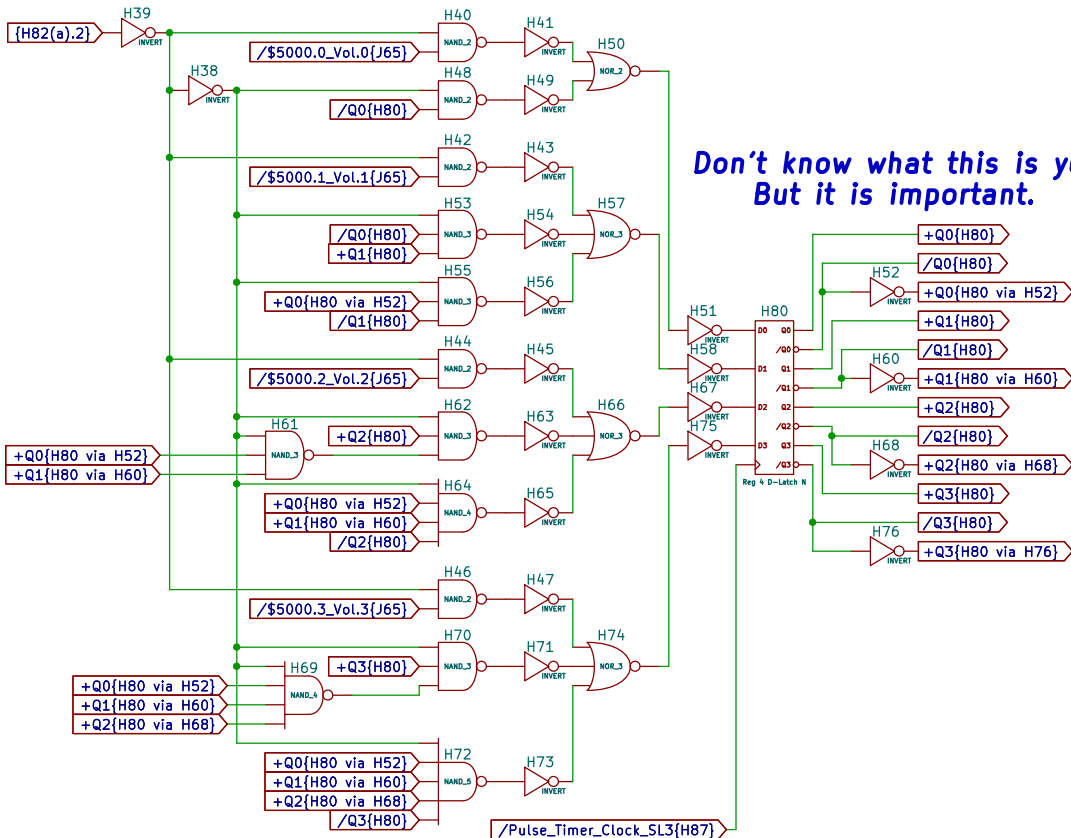


Note: The adder's logic is
verified correct in
Pulse_DAC_Adder_Logix.xlxs

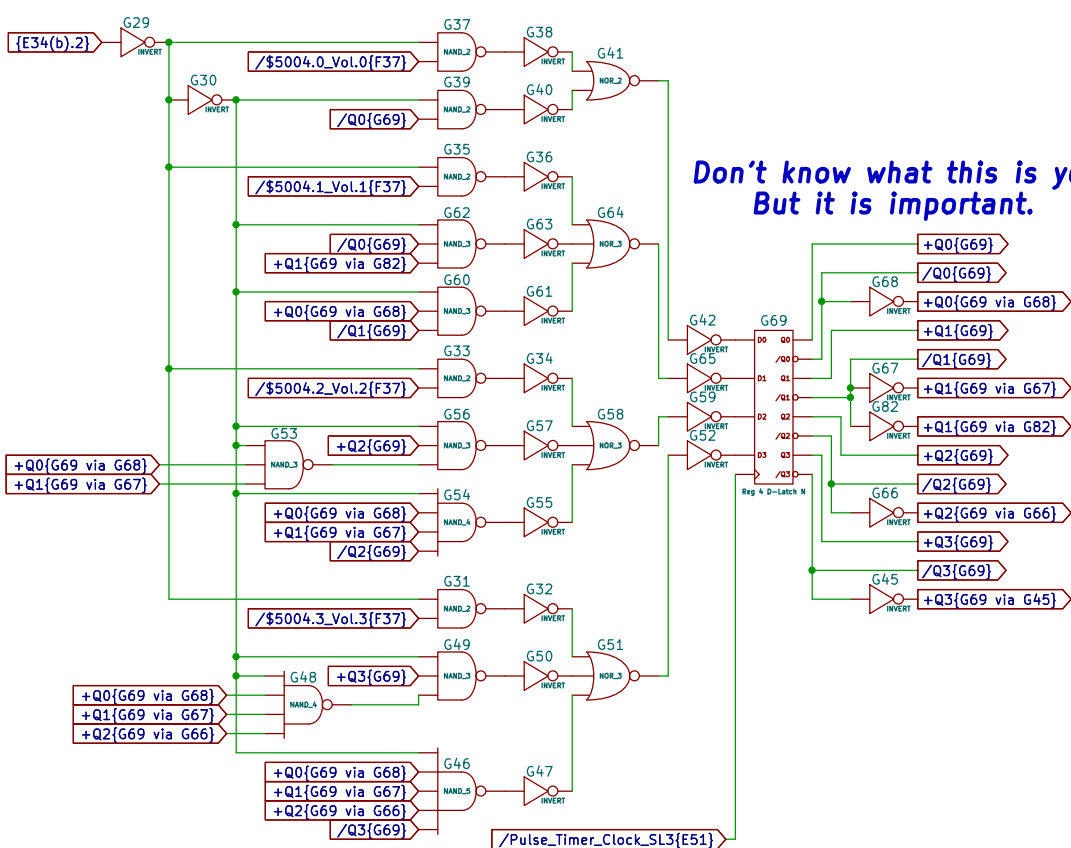
4-bit Adder Adding Pulse 1&2 DACs

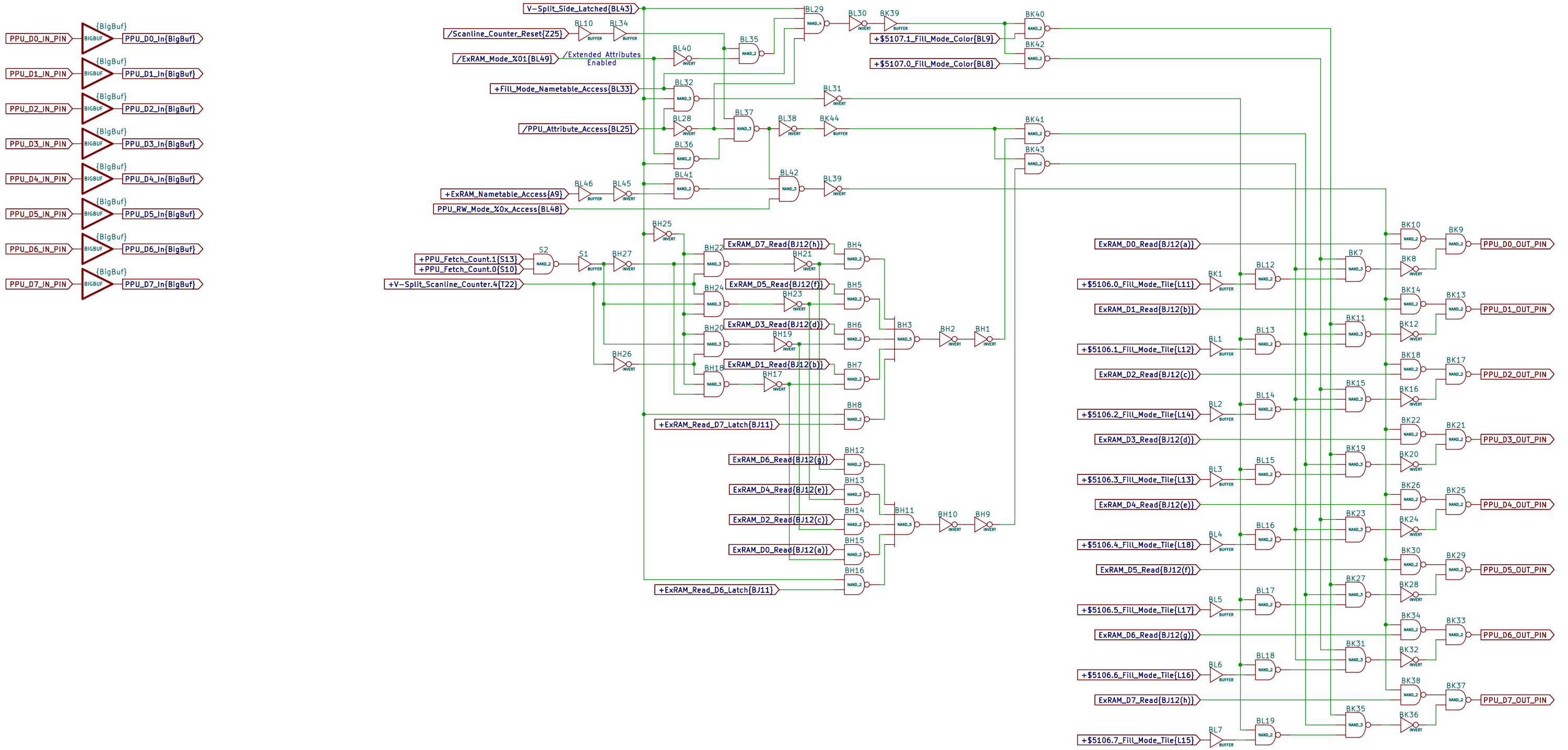


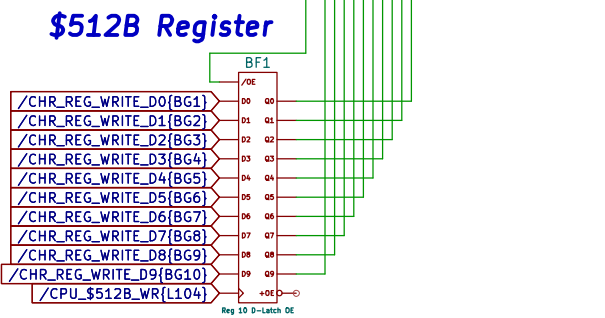
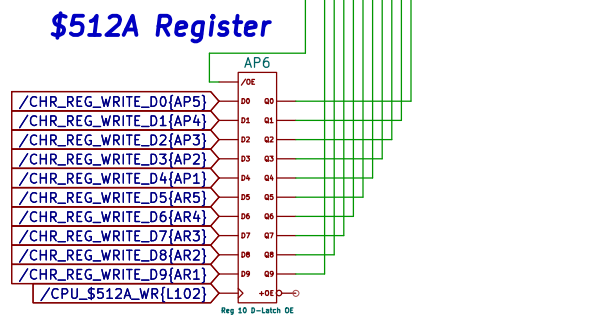
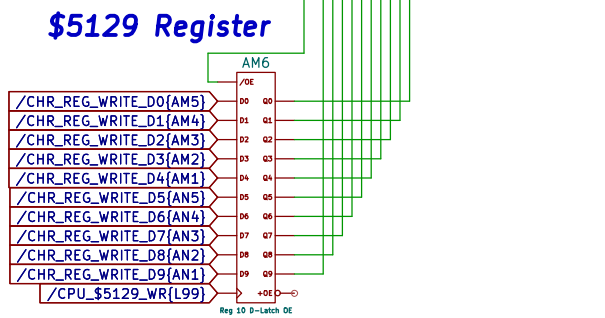
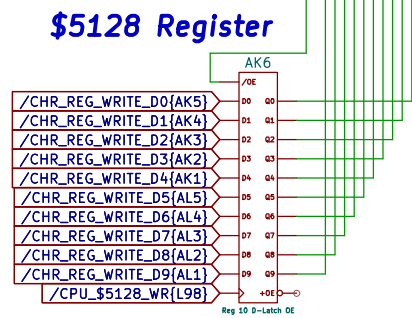
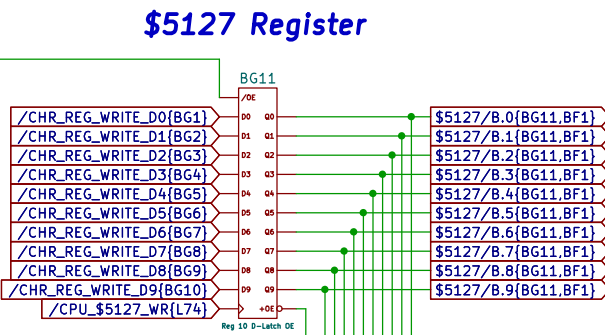
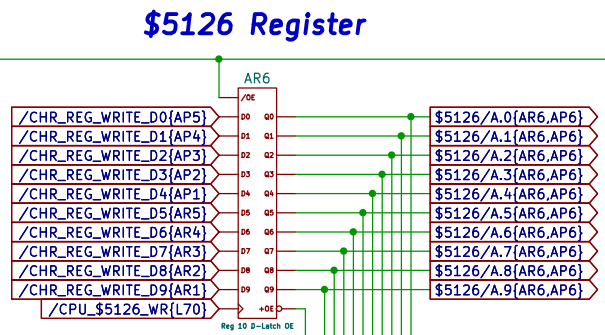
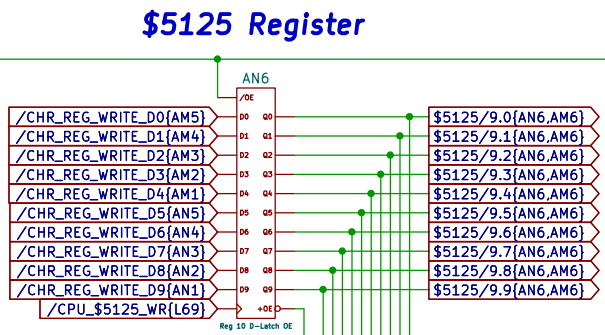
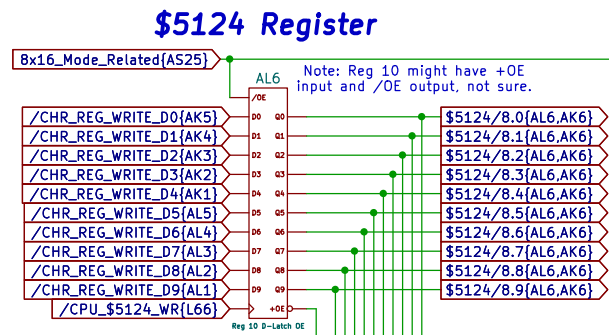
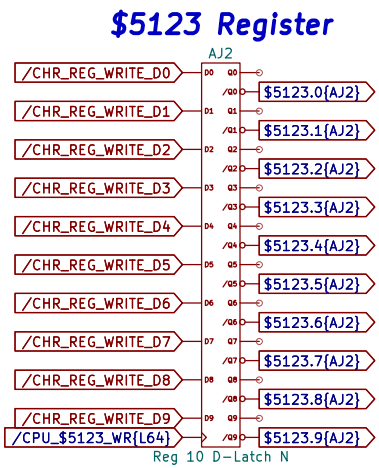
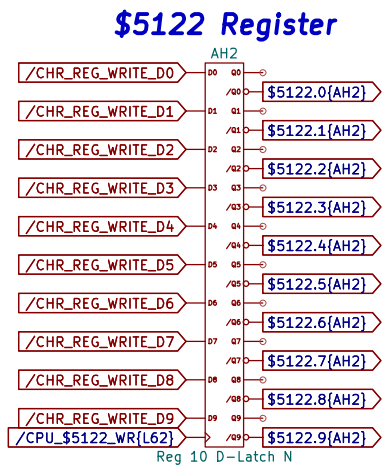
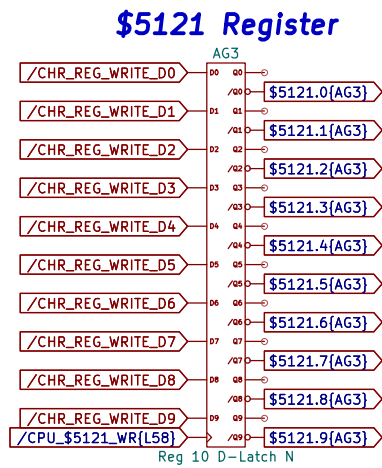
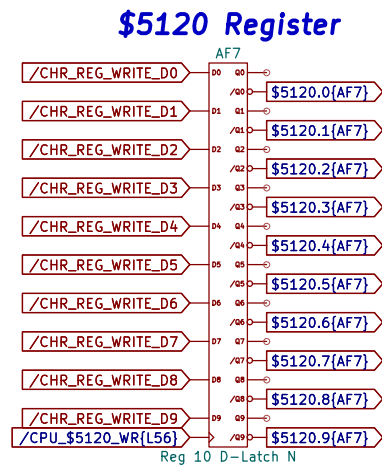
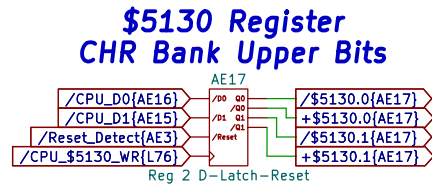
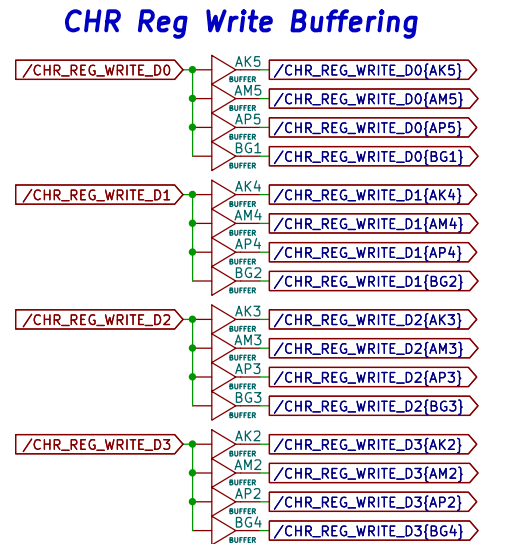
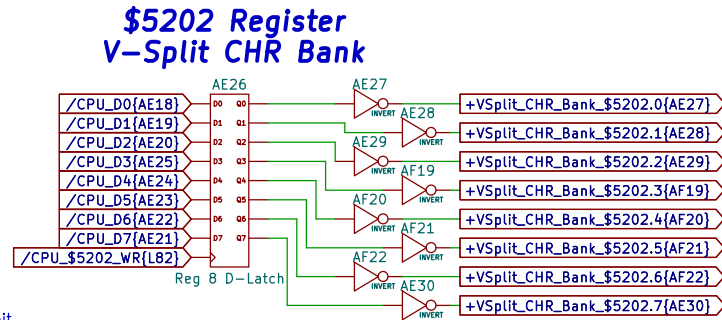
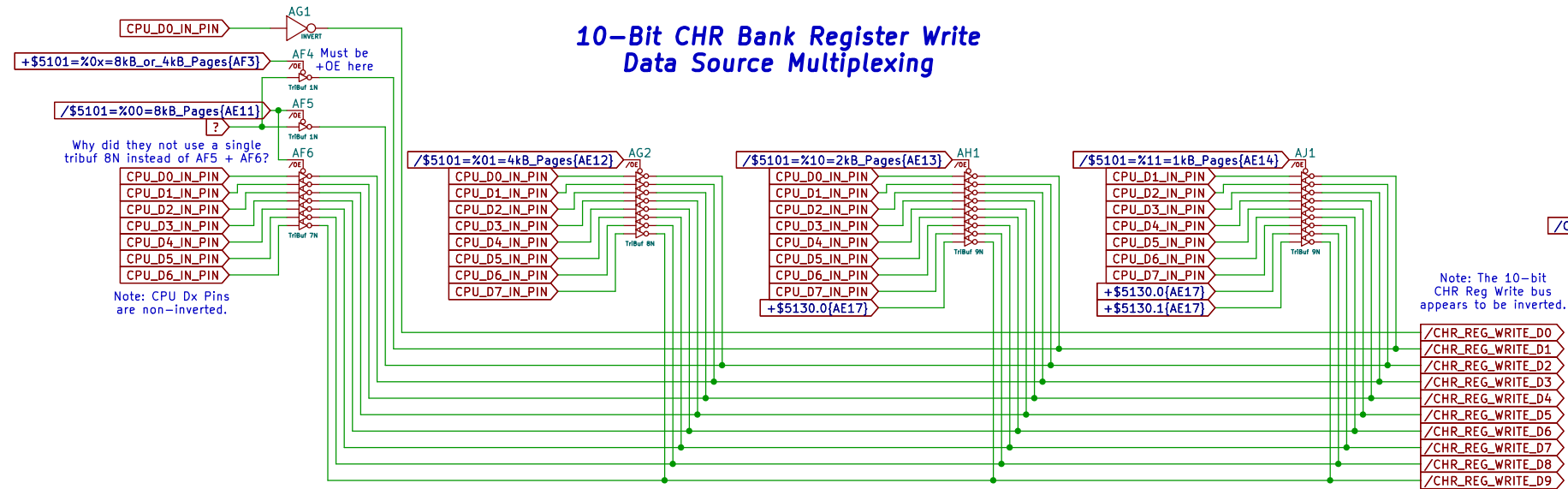
Don't know what this is yet
But it is important.

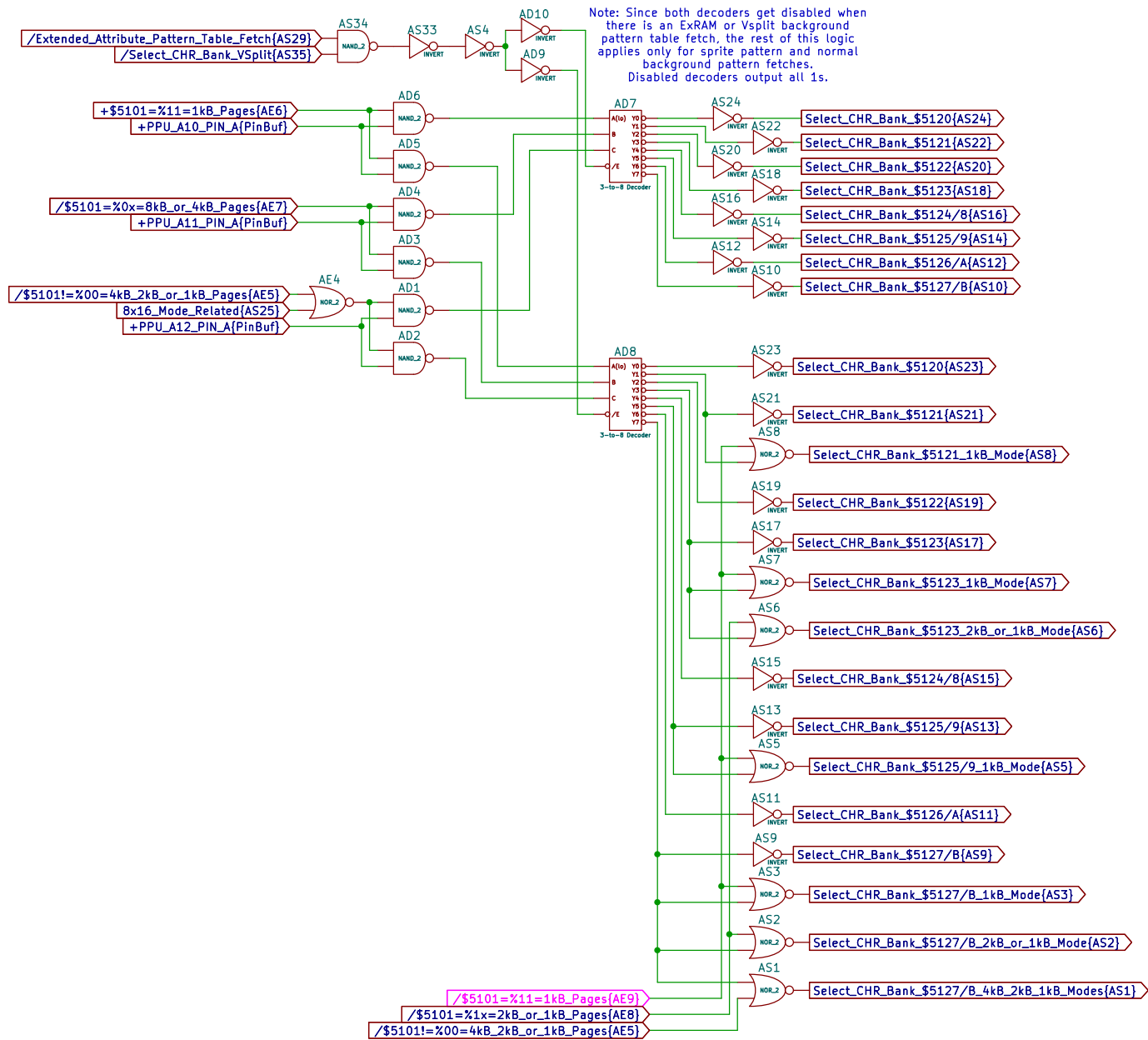


Don't know what this is yet
But it is important.

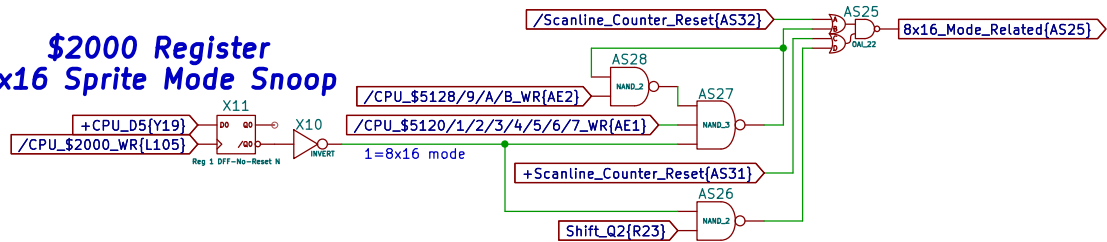




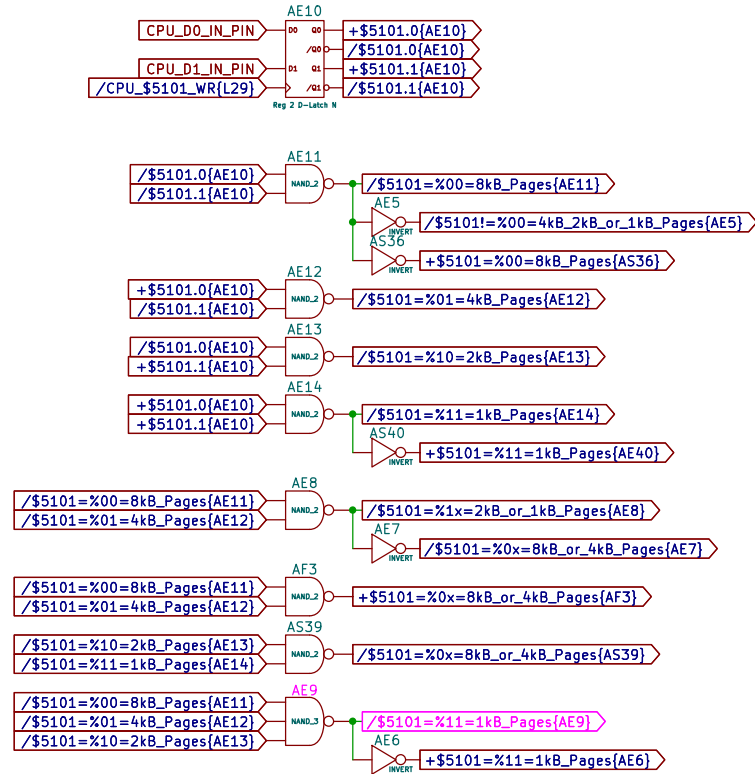




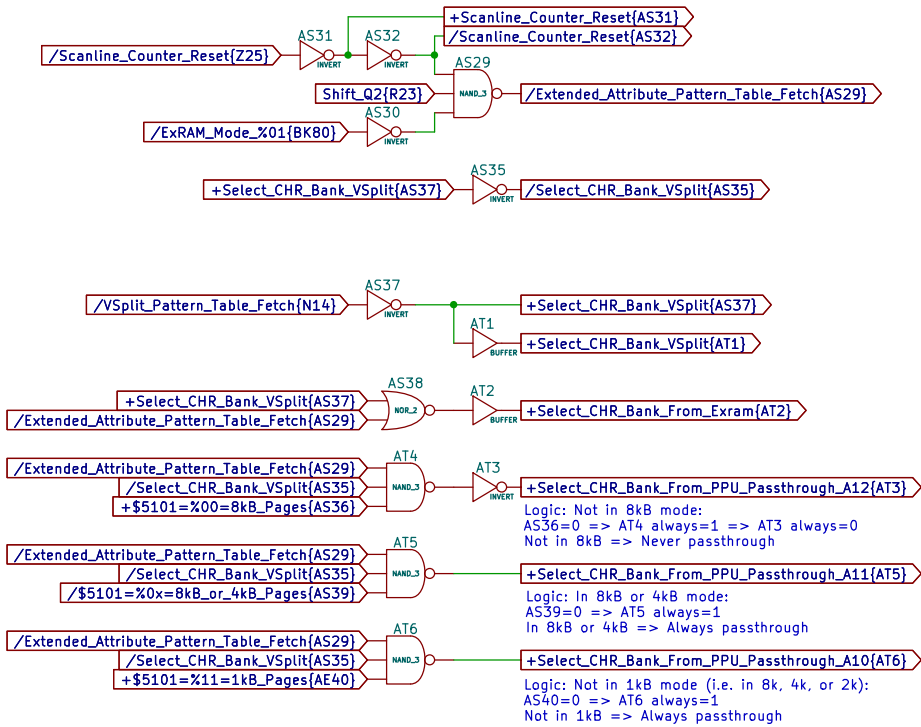
\$2000 Register 8x16 Sprite Mode Snoop



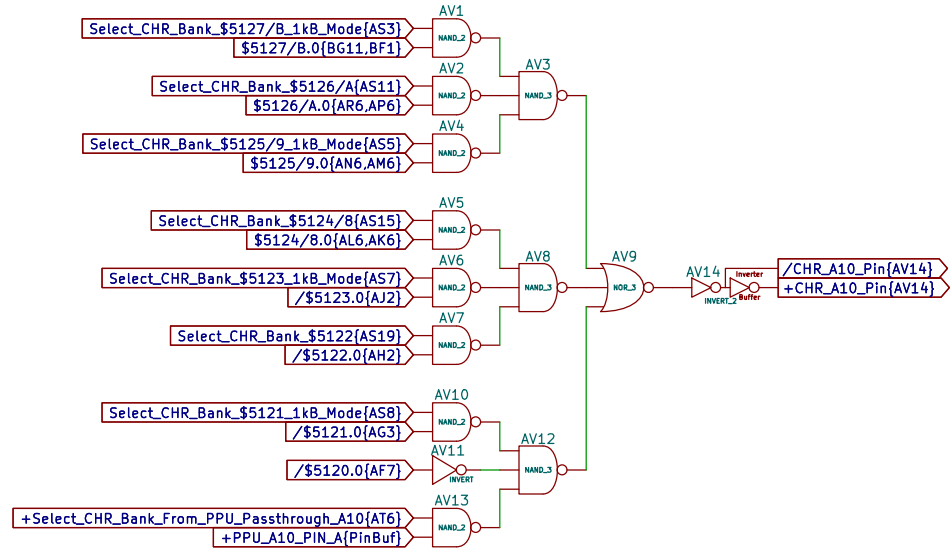
\$5101 Register CHR Mode



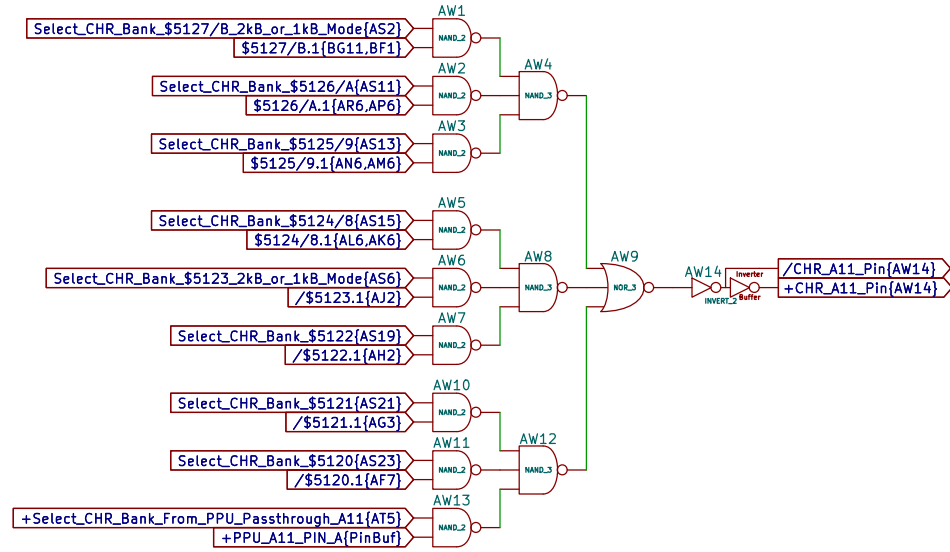
The logic in this area is strangely unoptimized.



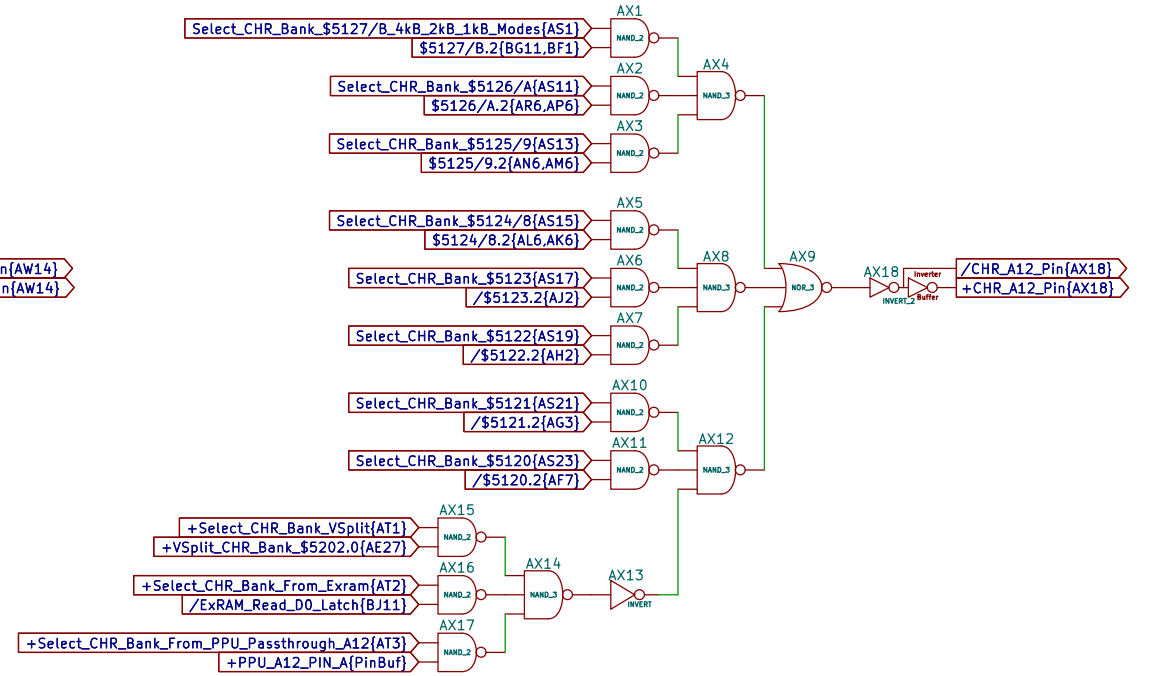
CHR A10



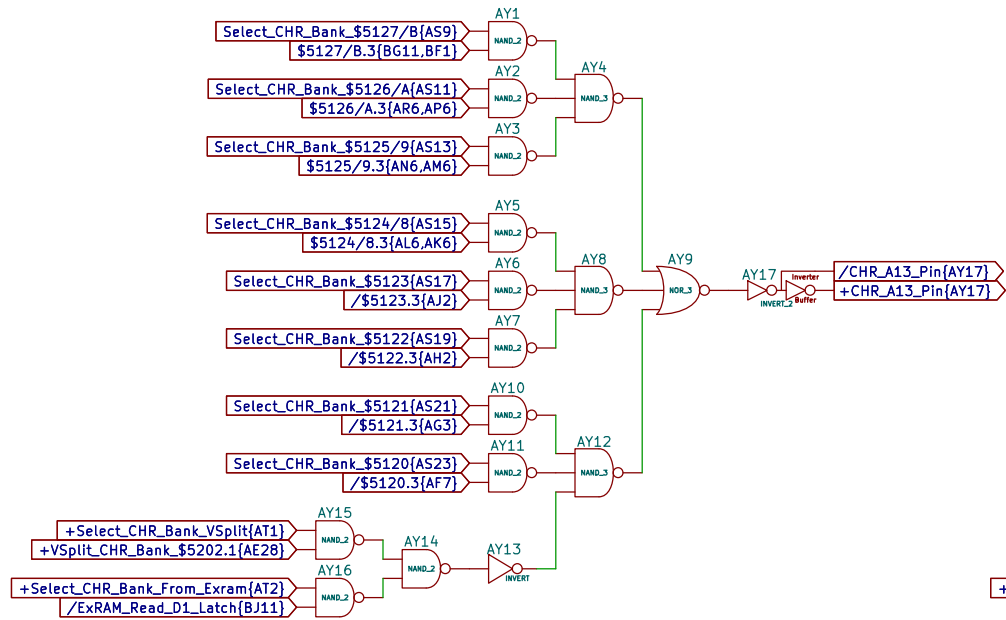
CHR A11



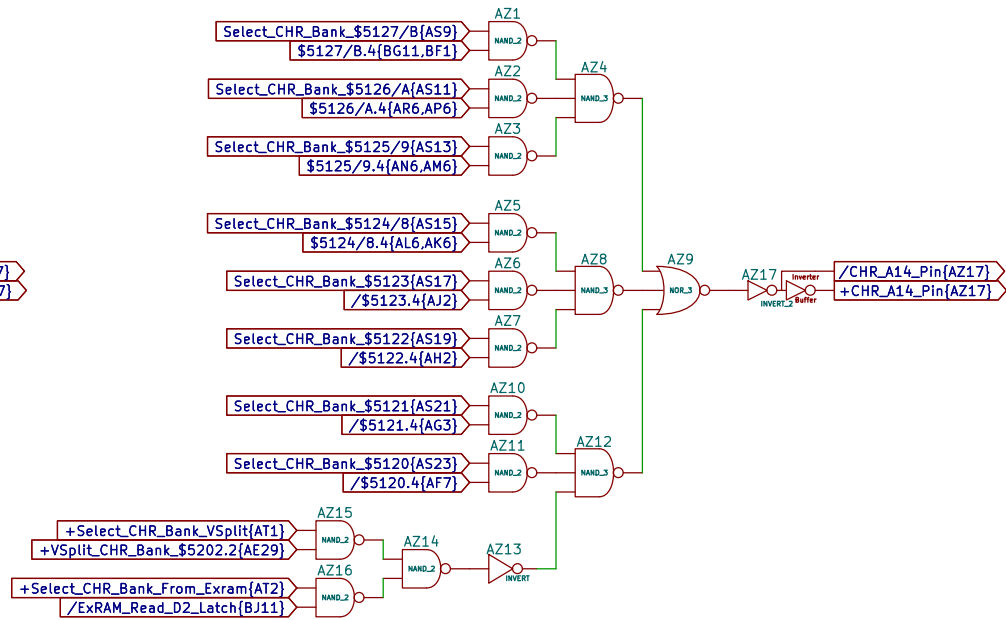
CHR A12



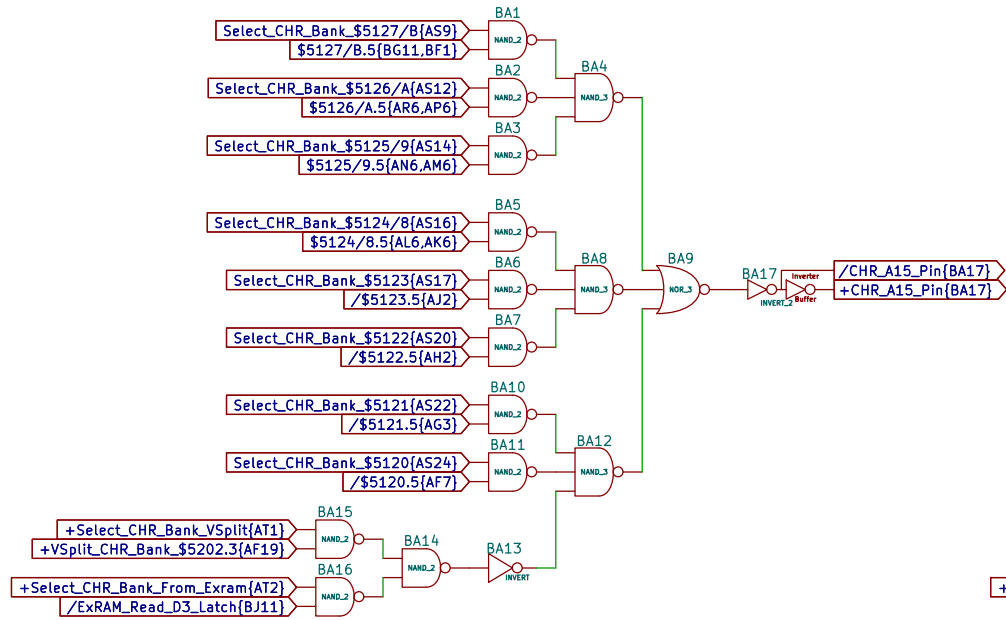
CHR A13



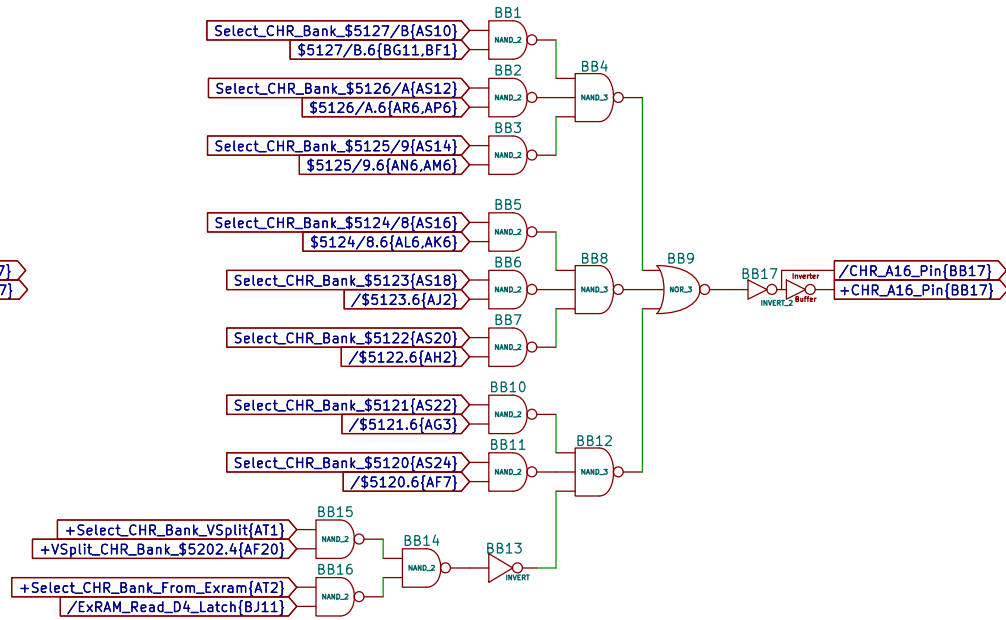
CHR A14



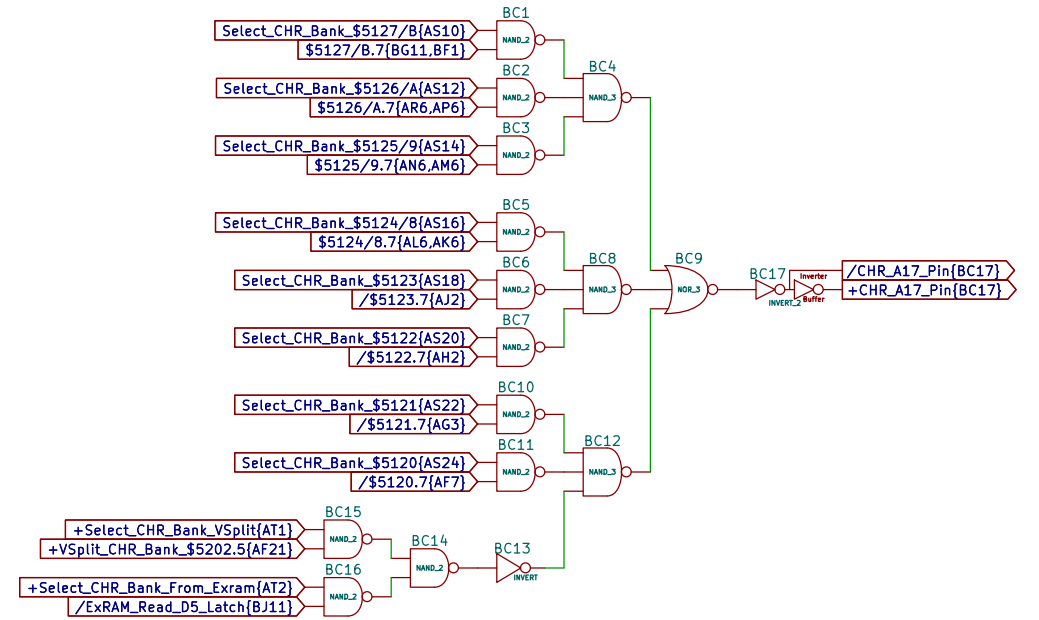
CHR A15



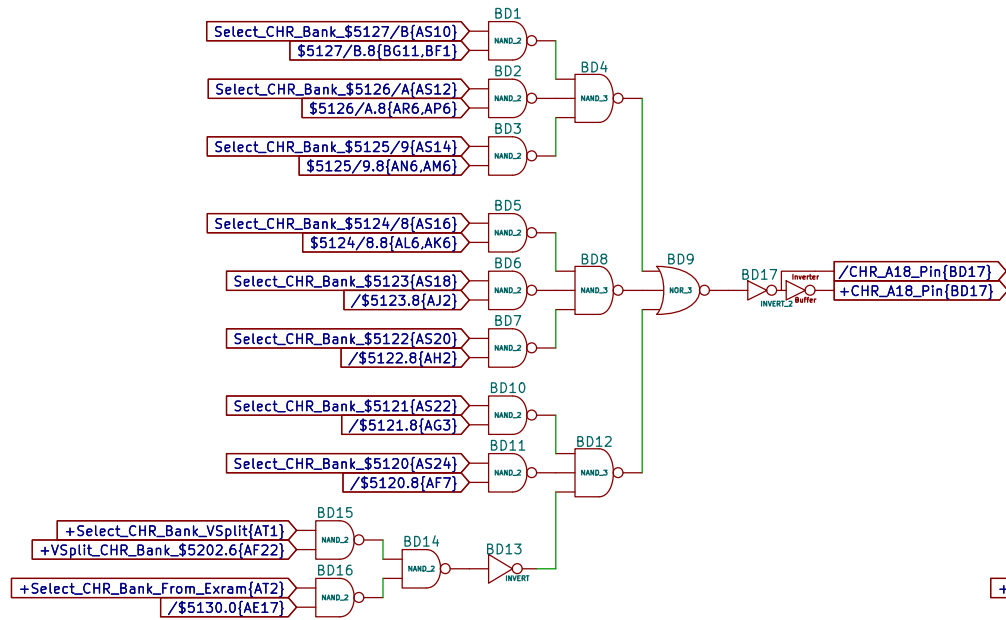
CHR A16



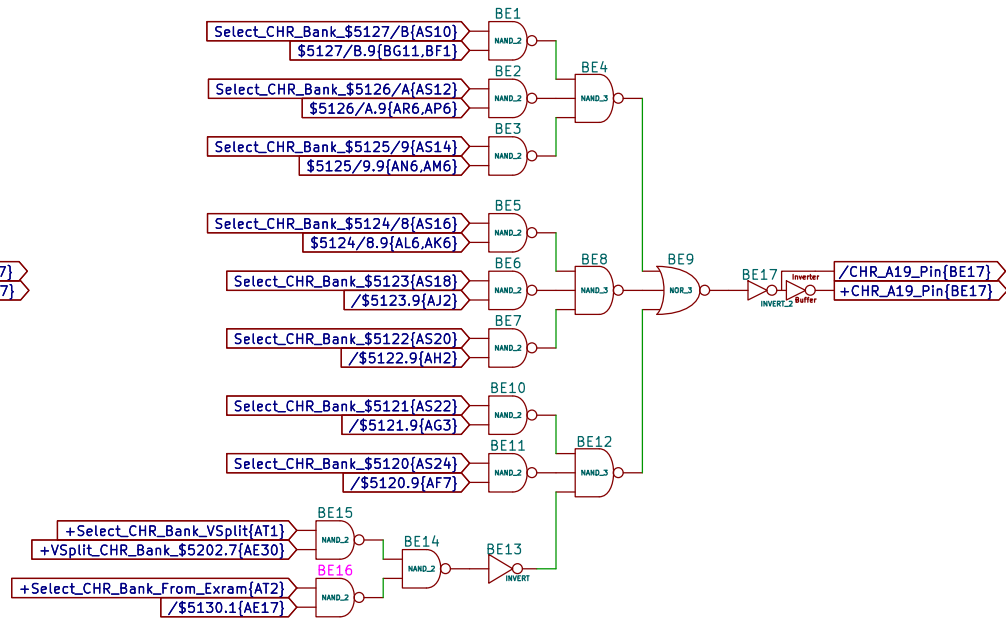
CHR A17



CHR A18



CHR A19



This sheet contains all residual items that don't serve any apparent purpose.

